

Earthquake Vulnerability of BC's Critical Infrastructure

by

Janos Toth

Vice President of Engineering
Quanta Energized Services



PACIFIC ENERGY INNOVATION ASSOCIATION

November 5, 2015
Vancouver BC, Canada



- Quanta Services Electric Power, Oil and Gas Pipe Lines
- Quanta Services is the largest outside electric service contractor in North America
- Preferred partner of leading utilities.
- From design and program management, installation and maintenance, energized services and emergency restoration,
- Quanta's expertise and resources offer a single source solution to any power infrastructure need.
- Founded in 1997
- Headquarter: Houston,Texas, USA
- Fortune 500 Company
- About 27,000 employees world wide (3,600 in Canada)
- Stock code: PWR



PACIFIC ENERGY INNOVATION ASSOCIATION



Some Canadian Operating Units:

- Allteck Line Contractors
- Valard Construction
- FRP Transmission Innovations
- Lex Engineering



Outline

- Critical Infrastructure
- Earthquake engineering terminology
- Earthquakes in BC
- Earthquake vulnerabilities of buildings, police, fire halls, schools, transportation, water, energy

Critical Infrastructure

- provides the essential services that underpin society and serve as the backbone of economy, security, and health.



Critical Infrastructure

- Commercial Facilities (offices, warehouses etc.)
- Housing, Offices, Schools
- Transportation (Roads, bridges, ports, railways, airports, tunnels)
- Communication facilities
- Information technology facilities
- Industrial facilities
- Energy delivery, Electrical, Natural Gas, Oil etc. lines
- Health facilities (hospitals etc.)
- Water and waste water facilities



PACIFIC ENERGY INNOVATION ASSOCIATION

Original Infrastructure Built for Earthquake

- Originally the majority of the BC's infrastructure was build without seismic (earthquake) considerations
- Simple it was unknown what seismic hazards existed in BC
- The Subduction Earthquake affects the West Coast of North America

Insurance Bureau of Canada Study 1

- a 9.0 magnitude earthquake off the western coast of Vancouver Island would have a major impact on the economy.
- It estimates that scenario would cause \$62 billion in direct damage in the region, primarily due to building damage, and \$12.7 billion in indirect impact caused by things like supply chain interruption and infrastructure damage.

Insurance Bureau of Canada Study 2

- The report does not look at loss of life or casualties, focusing solely on the impact to the economy, but says it considers an earthquake "sufficiently threatening and devastating to warrant prudent planning and preparation now."

<http://www.cbc.ca/news/business/major-earthquake-could-cause-75b-in-damage-study-warns-1.2287121>

Ref: Insurance Bureau of Canada Study 2010

November 4, 2015

Building Code

- Building Code is designed to minimize loss of life
- It does not mean that structures will be undamaged and/or fully functional after the design earthquake.



PACIFIC ENERGY INNOVATION ASSOCIATION

Earthquake Terminology

- Life Lines
- Return Period
- Earthquake Risk
- Peak Ground Acceleration (PGA)
- Liquefaction



PACIFIC ENERGY INNOVATION ASSOCIATION

Life Lines

- Lifelines are structures that are important or critical for a community to function, such as roadways, pipelines, power lines, sewers, communications, and port facilities.

Return Period

- Reoccurrence interval (repeat interval) is an estimated likelihood of an event, such as earthquake, flood, landslide, etc.) to occur.
- For example 1 in 50 year (probability happening $1/50=0.02$ or 2 %
- 1 in 100 $1/100=0.01$ or 1%
- 1 in 475 $1/475=0.002$
- 1 in 2,745 $1/2745=0.0004$
- 1 in 10,000 $1/10,000=0.0001$



PACIFIC ENERGY INNOVATION ASSOCIATION

Earthquake risk

- Risk = Probability x consequence

Inspection Priority Categories

	1	11	7	4	2	1
	2	16	13	8	6	3
	3	20	17	14	9	5
	4	23	21	18	15	10
	5	25	24	22	19	12
Probability Categories		E	D	C	B	A
		Consequence Categories				

Peak Ground Acceleration (PGA)

- Horizontal earthquake acceleration on the ground

0.001 g (0.01 m/s^2) – perceptible by people

0.02 g (0.2 m/s^2) – people lose their balance

0.50 g – very high;



PACIFIC ENERGY INNOVATION ASSOCIATION

Earthquake Magnitude, Ground Peak Acceleration

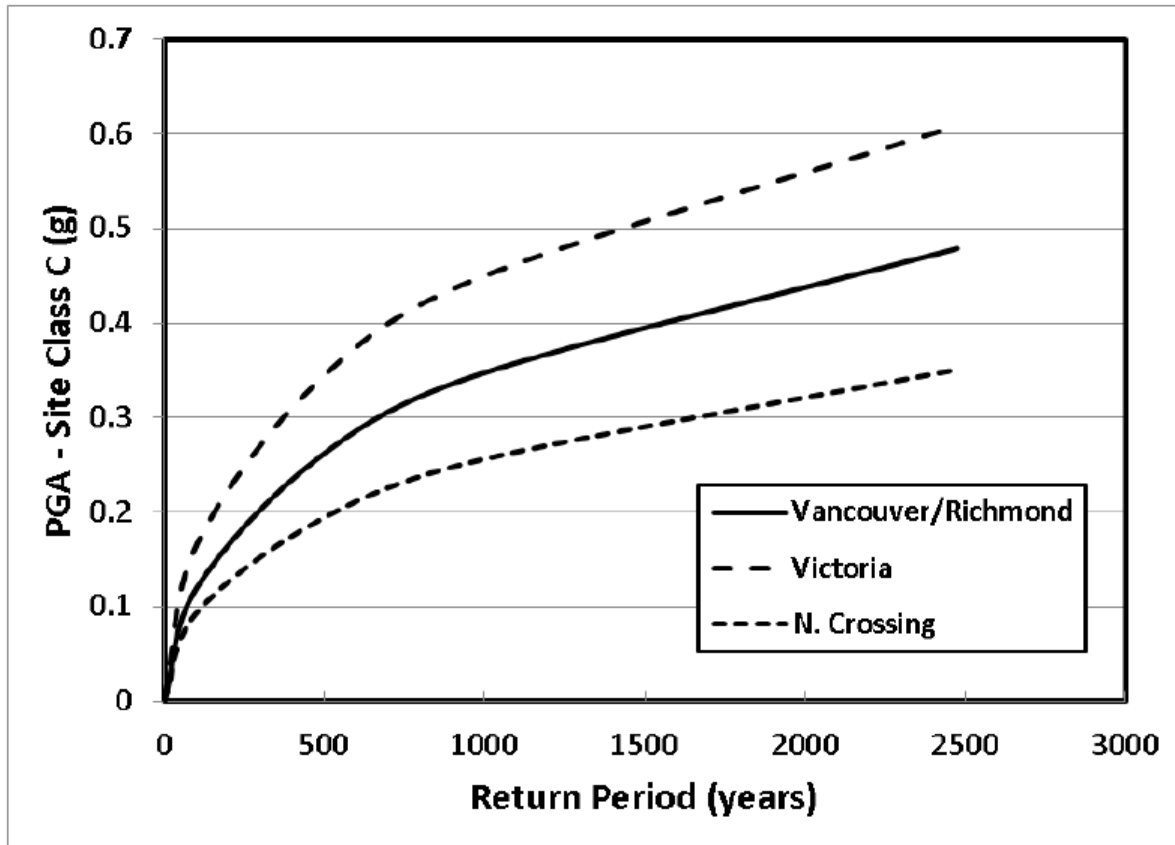
Instrumental Intensity	Acceleration (g)	Velocity (cm/s)	Perceived Shaking	Potential Damage
I	< 0.0017	< 0.1	Not felt	None
II-III	0.0017 - 0.014	0.1 - 1.1	Weak	None
IV	0.014 - 0.039	1.1 - 3.4	Light	None
V	0.039 - 0.092	3.4 - 8.1	Moderate	Very light
VI	0.092 - 0.18	8.1 - 16	Strong	Light
VII	0.18 - 0.34	16 - 31	Very strong	Moderate
VIII	0.34 - 0.65	31 - 60	Severe	Moderate to heavy
IX	0.65 - 1.24	60 - 116	Violent	Heavy
X+	> 1.24	> 116	Extreme	Very heavy

Magnitude	Typical Maximum Modified Mercalli Intensity
1.0 - 3.0	I
3.0 - 3.9	II - III
4.0 - 4.9	IV - V
5.0 - 5.9	VI - VII
6.0 - 6.9	VII - IX
7.0 and higher	VIII or higher

Ref: United States Geological Survey http://earthquake.usgs.gov/learn/topics/mag_vs_int.php

November 4, 2015

PGA vc Return Period



Ref: Natural Resources Canada 2013

November 4, 2015

Peak Ground Acceleration



<http://bcndpcaucus.ca/news/liberals-still-apathetic-on-emergency-response/>

Liquefaction

- A process by which water-saturated sediment temporarily loses strength and acts as a fluid, like when you wiggle your toes in the wet sand near the water at the beach. This effect can be caused by earthquake shaking.

<http://earthquake.usgs.gov/learn/glossary/?term=liquefaction>

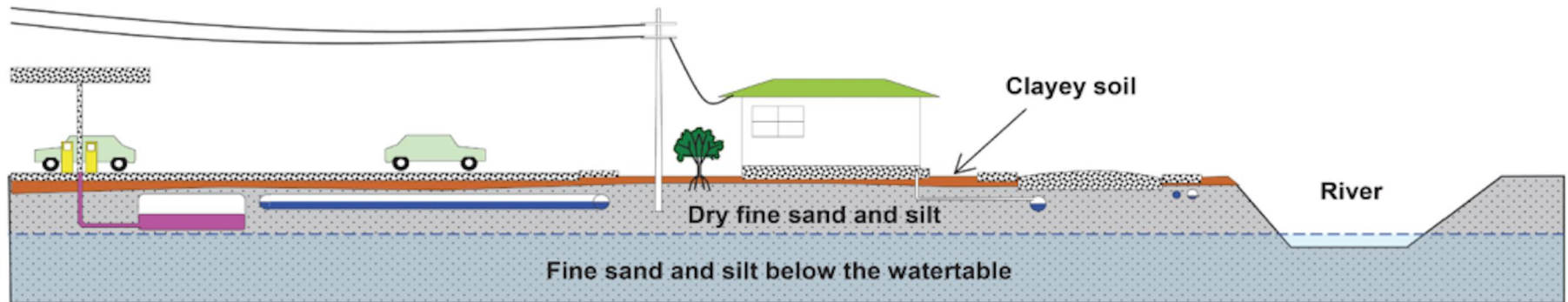


PACIFIC ENERGY INNOVATION ASSOCIATION

Liquefaction and its Effects

Before the Earthquake

Areas of flat, low lying land with groundwater only a few metres below the surface, can support buildings and roads, buried pipes, cables and tanks under normal conditions.



During and after the Earthquake

During the earthquake fine sand, silt and water moves up under pressure through cracks and other weak areas to erupt onto the ground surface. Near rivers the pressure is relieved to the side as the ground moves sideways into the river channels.

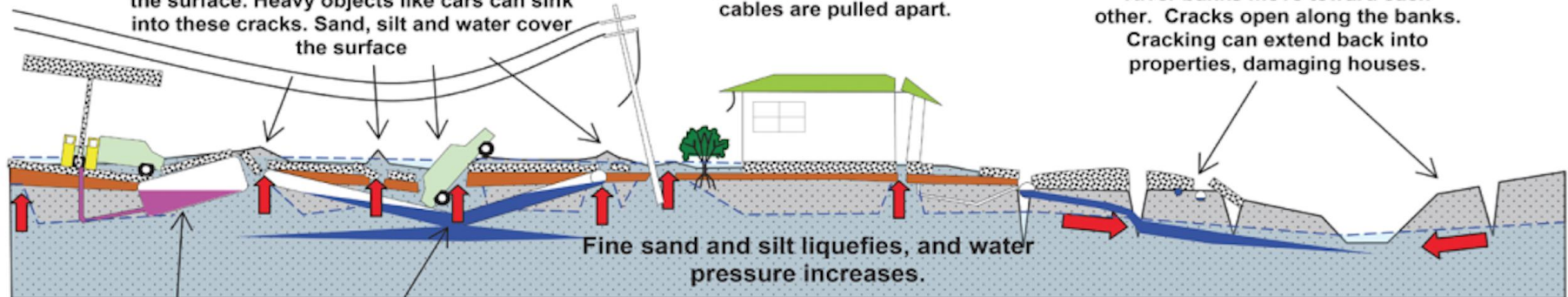
Sand Boils (Sand Volcanoes)

Sand, silt and water erupts upward under pressure through cracks and flows out onto the surface. Heavy objects like cars can sink into these cracks. Sand, silt and water cover the surface

Power poles are pulled over by their wires as they can't be supported in the liquefied ground. Underground cables are pulled apart.

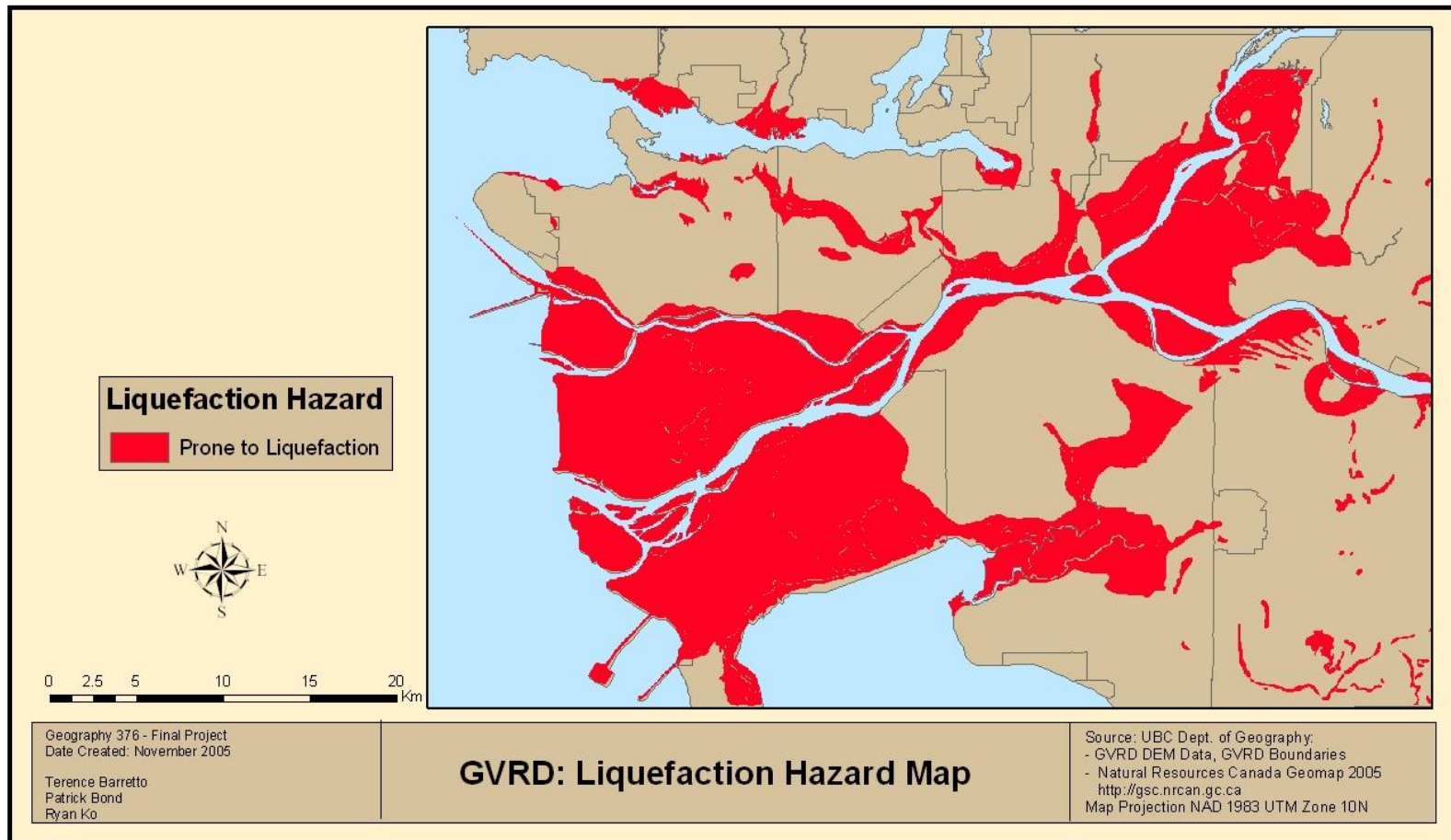
Lateral Spreading

River banks move toward each other. Cracks open along the banks. Cracking can extend back into properties, damaging houses.

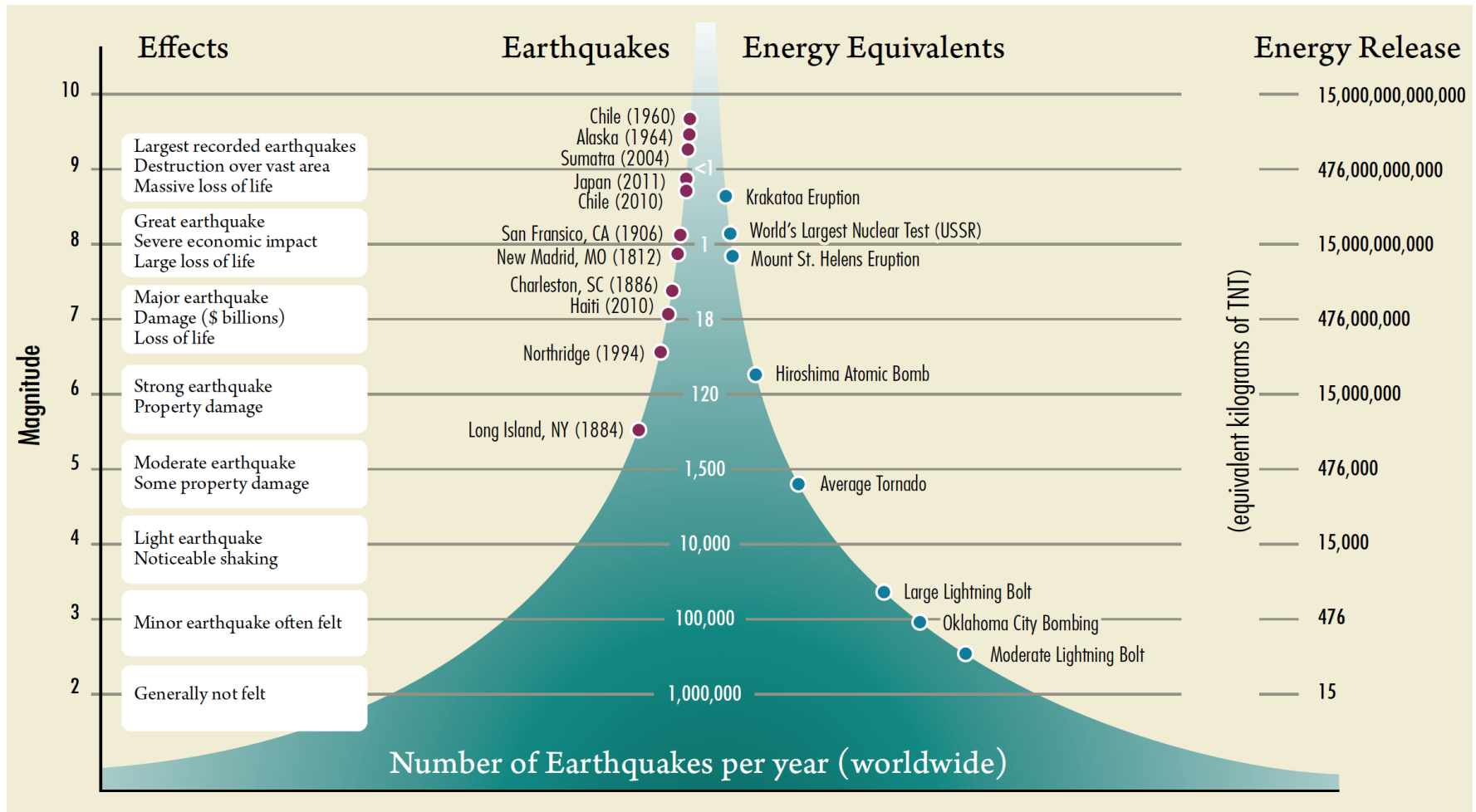


Tanks, pipes and manholes float up in the liquefied ground and break through the surface. Pipes break, water and sewage leaks into the ground.

Liquefaction Hazard Map



Earthquake Statistics



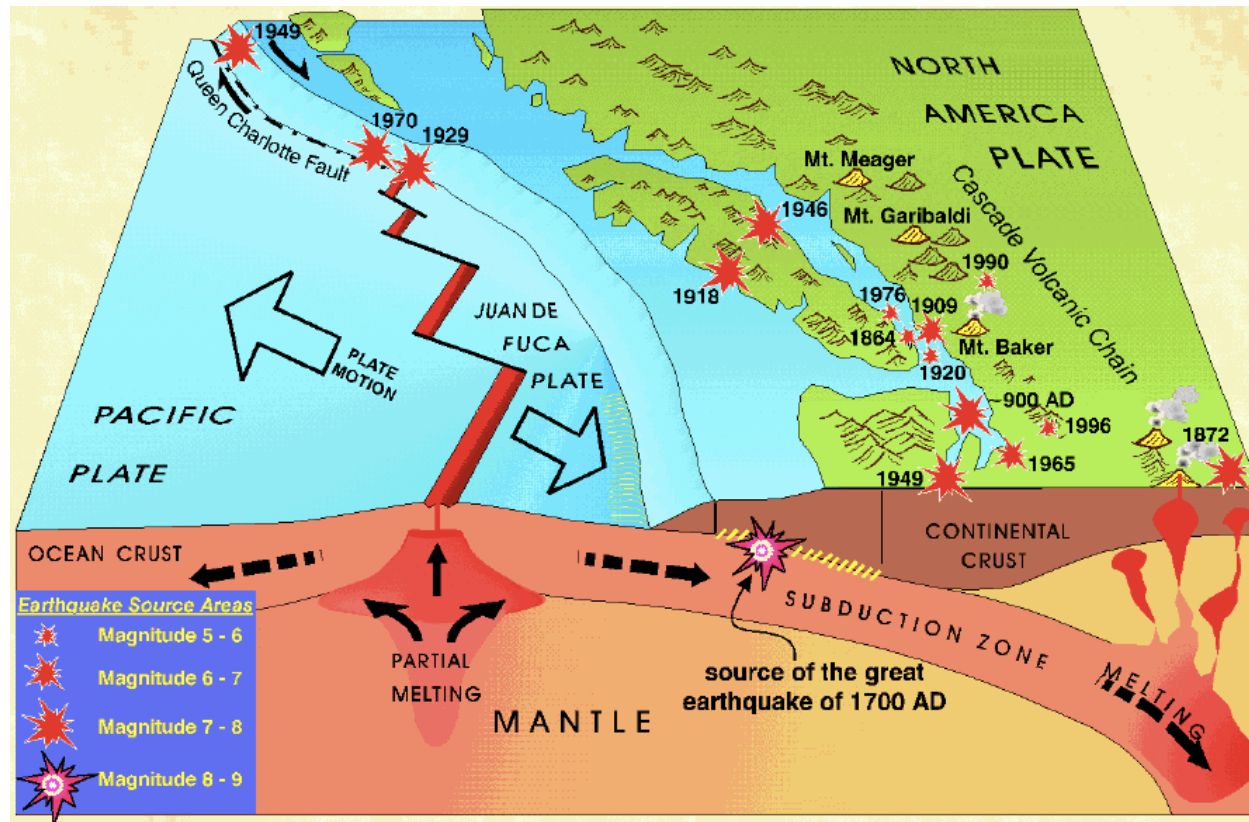
Richter Scale and Category

Category		Effects	Richter Scale
I.	Instrumental	Not felt	1-2
II.	Just Perceptible	Felt by only a few people on upper floors of tall buildings	3
III.	Slight	Felt by people lying down, seated on hard surfaces, or in the upper floors of tall buildings	3.5
IV.	Perceptible	Felt indoors by many, by few outside, and windows rattle	4
V.	Rather Strong	Generally felt by everyone; sleeping people may be awakened	4.5
VI.	Strong	Trees sway, chandeliers swing, bells ring, some damage from falling objects	5
VII.	Very Strong	General alarm; walls and plaster crack	5.5
VIII.	Destructive	Felt in moving vehicles; chimneys collapse; poorly constructed buildings seriously damaged	6
IX.	Ruinous	Some houses collapse; pipes break	6.5
X.	Disastrous	Obvious ground cracks; railroad tracks bent; some landslides on steep hills	7
XI.	Very Disastrous	Few buildings survive; bridges damaged or destroyed; all services interrupted (electrical, water, sewage, railroad); severe landslides	7.5
XII.	Catastrophic	Total destruction; objects thrown into the air; river courses and topography altered	8

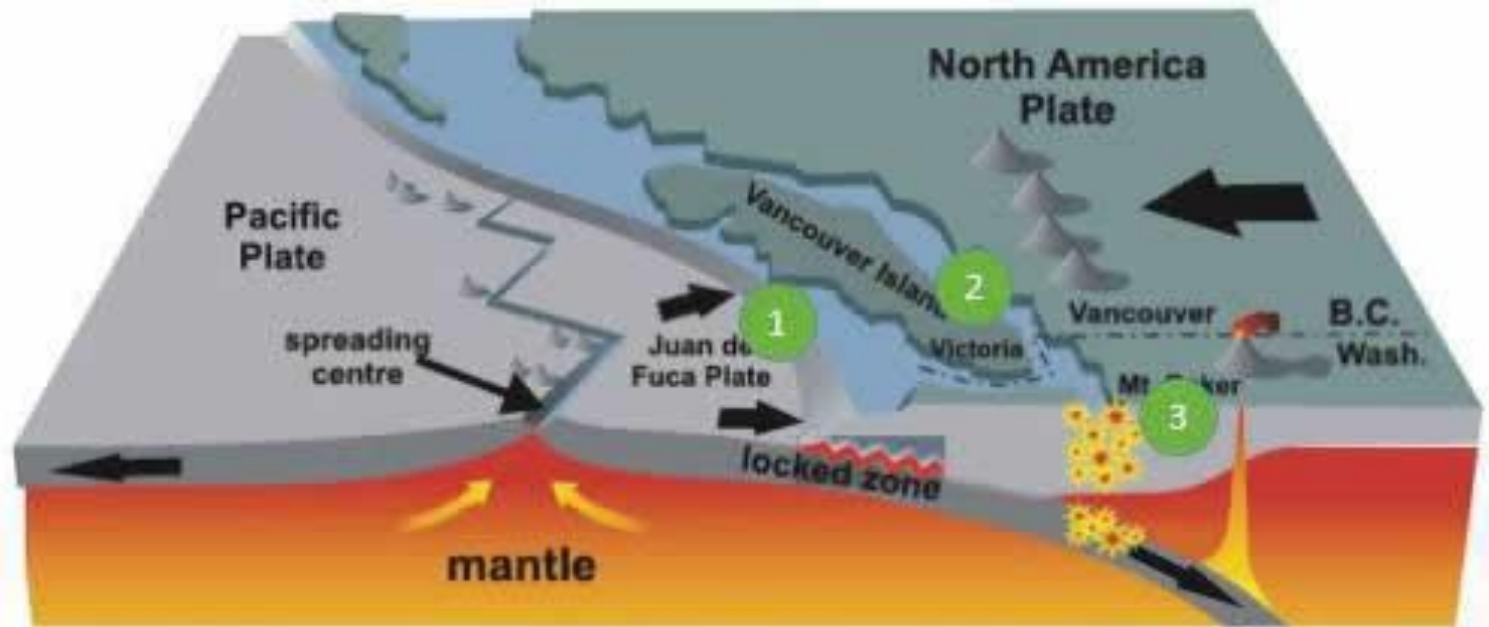
Megathrust Earthquake

- In the Cascadia subduction zone 13 megathrust events have been identified in the last 6000 years,
- An average one every 500 to 600 years.
- However, they have not happened regularly. Some have been as close together as 200 years and some have been as far apart as 800 years.
- The last one was about 300 years ago. January 26, 1700
Based on written record from Japan.

Seismic Geography



Expected Large Earthquakes in BC

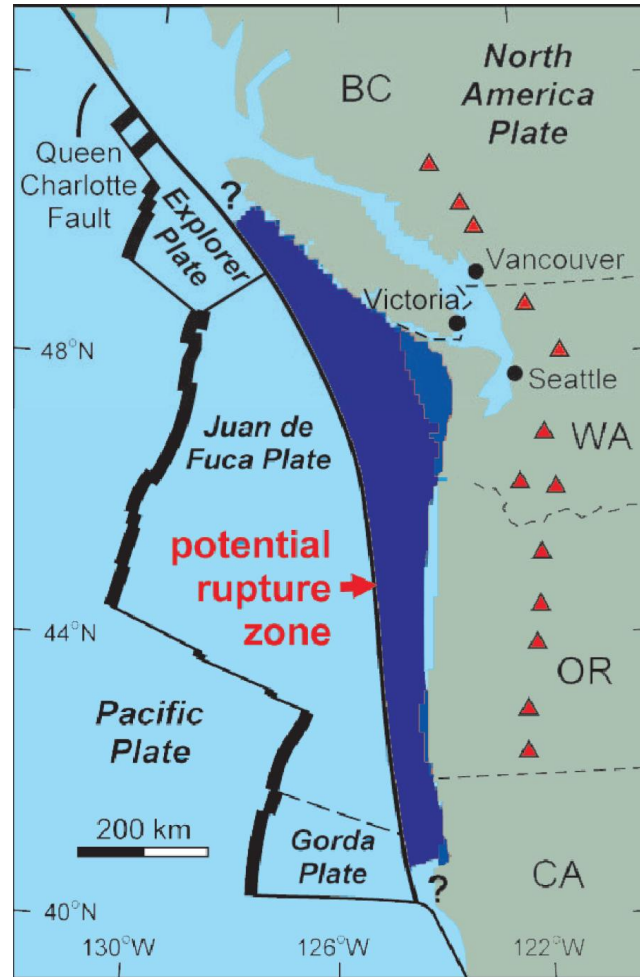


(Image credit: John Cague/MRCM)

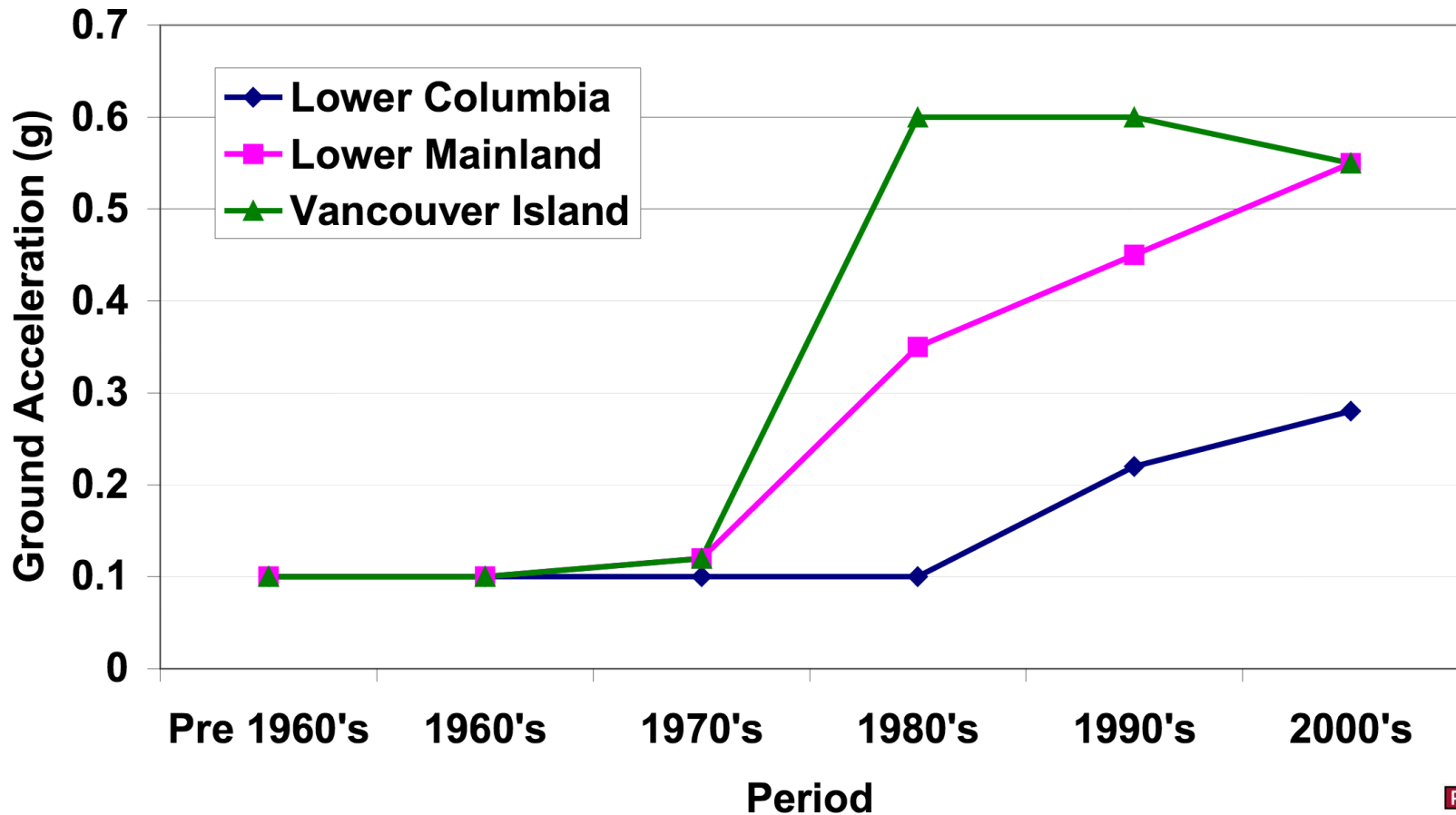
1. Cascadia Subduction Zone ("megathrust") earthquake - M9
2. Georgia Strait: shallow crustal earthquake - M 7.3
3. Deep, intraslab earthquake - M 6.8

<http://vancouver.ca/images/cov/content/earthquake-plates.jpg>

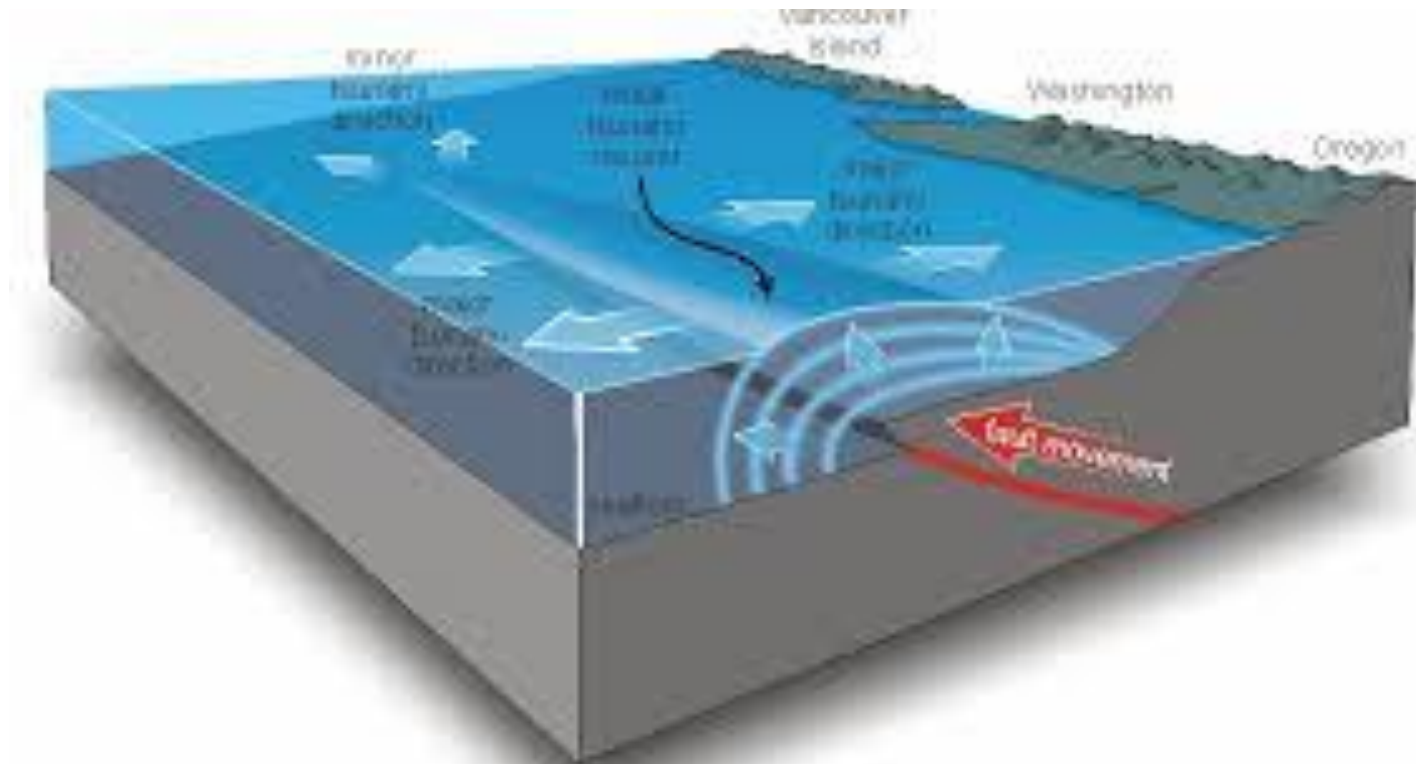
Potential Rupture Zone of the Cascadia Subduction Zone



Ground Acceleration Change in Knowledge



Earthquake Tsunami

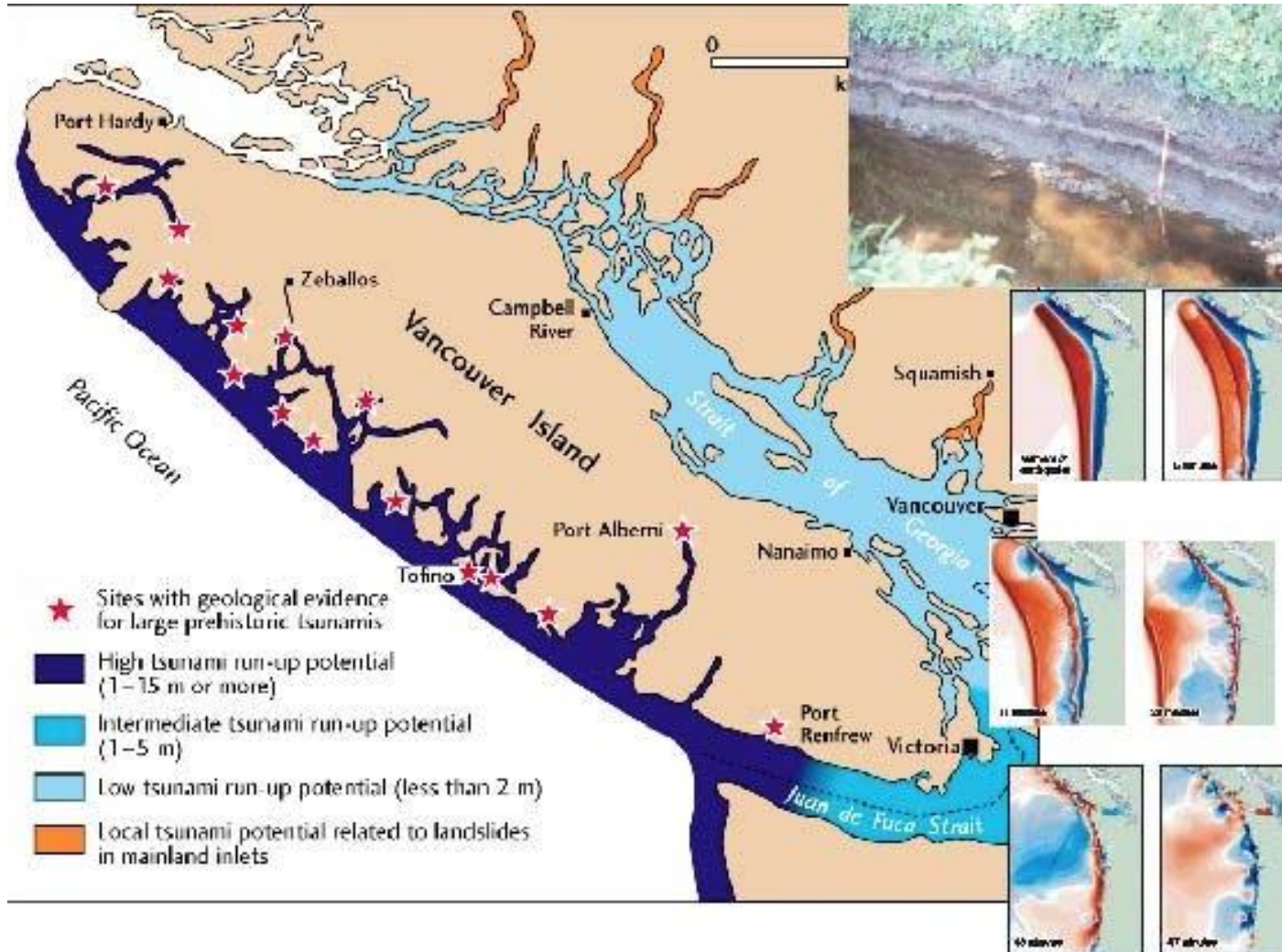


https://encrypted-tbn2.gstatic.com/images?q=tbn:ANd9GcRXwQzgMWKKIXPM-KKXDHuP2sR07H0rEUbewkb_h4G-_148Gs1F



PACIFIC ENERGY INNOVATION ASSOCIATION

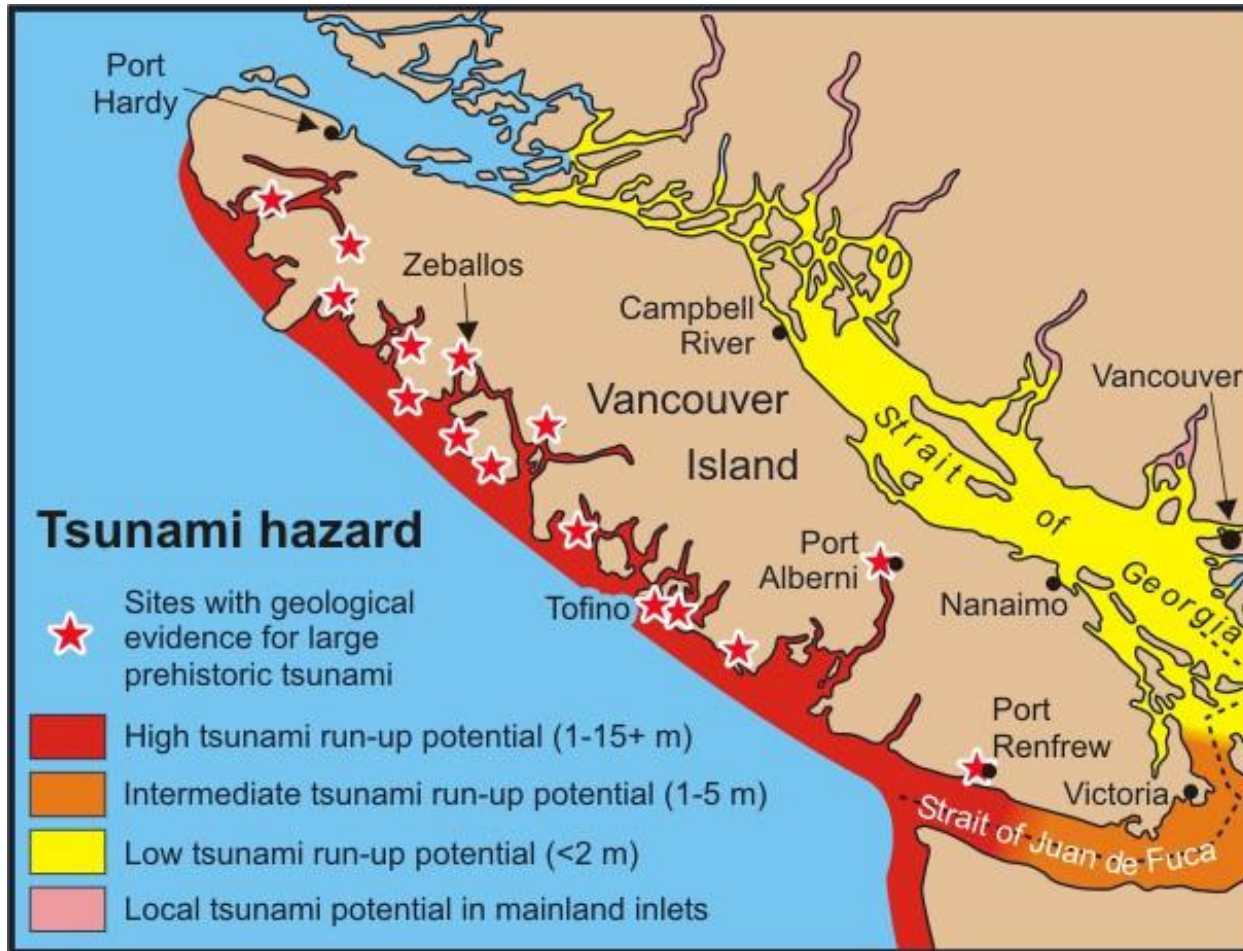
Tsunami



Tsunami Landing Times

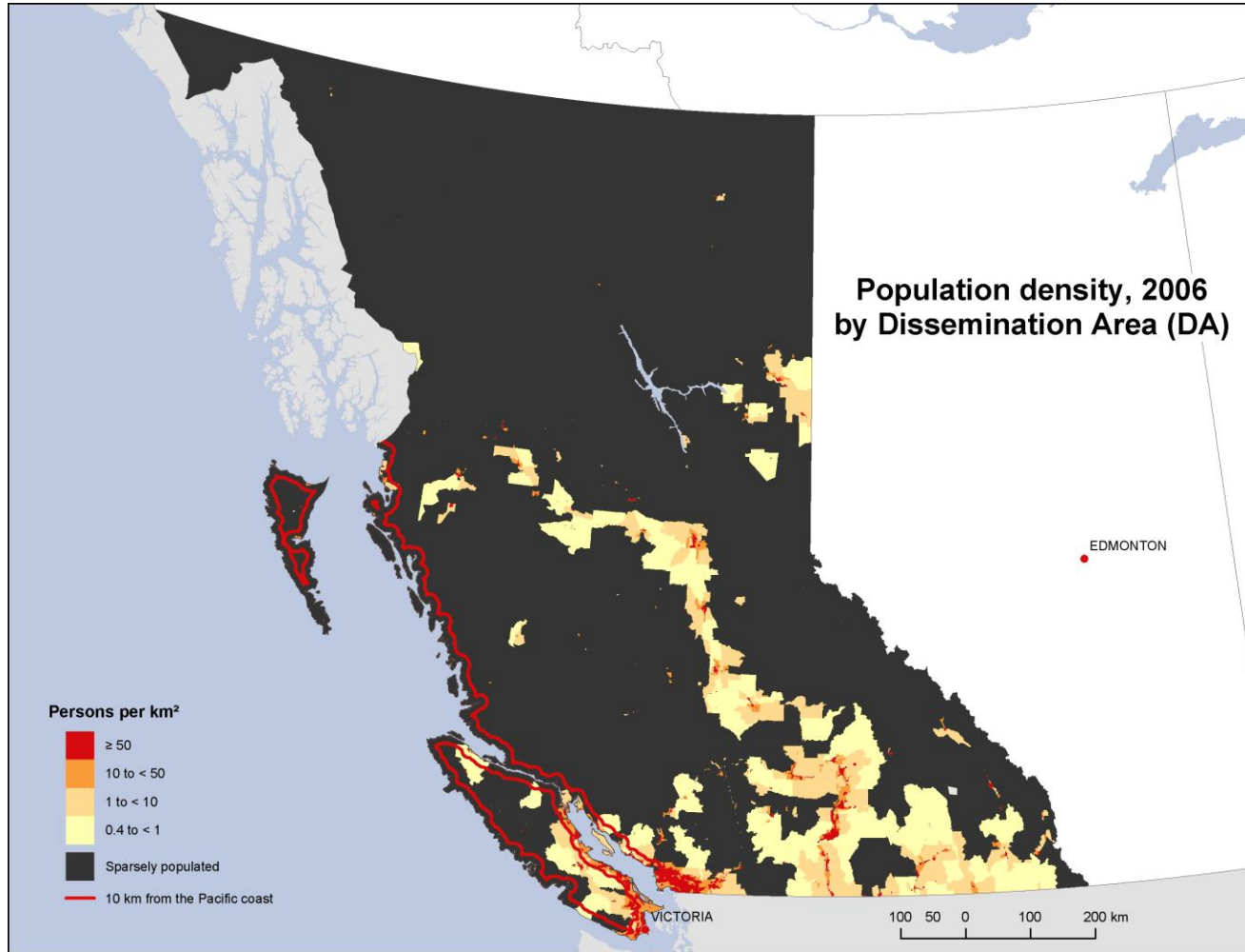
- West coast of Vancouver Island (10 to 20 min)
- Victoria (2 to 4 m within 75 minutes)
- Port Alberni (10 to 20 minutes 20 to 40 m)

Tsunami Hazard



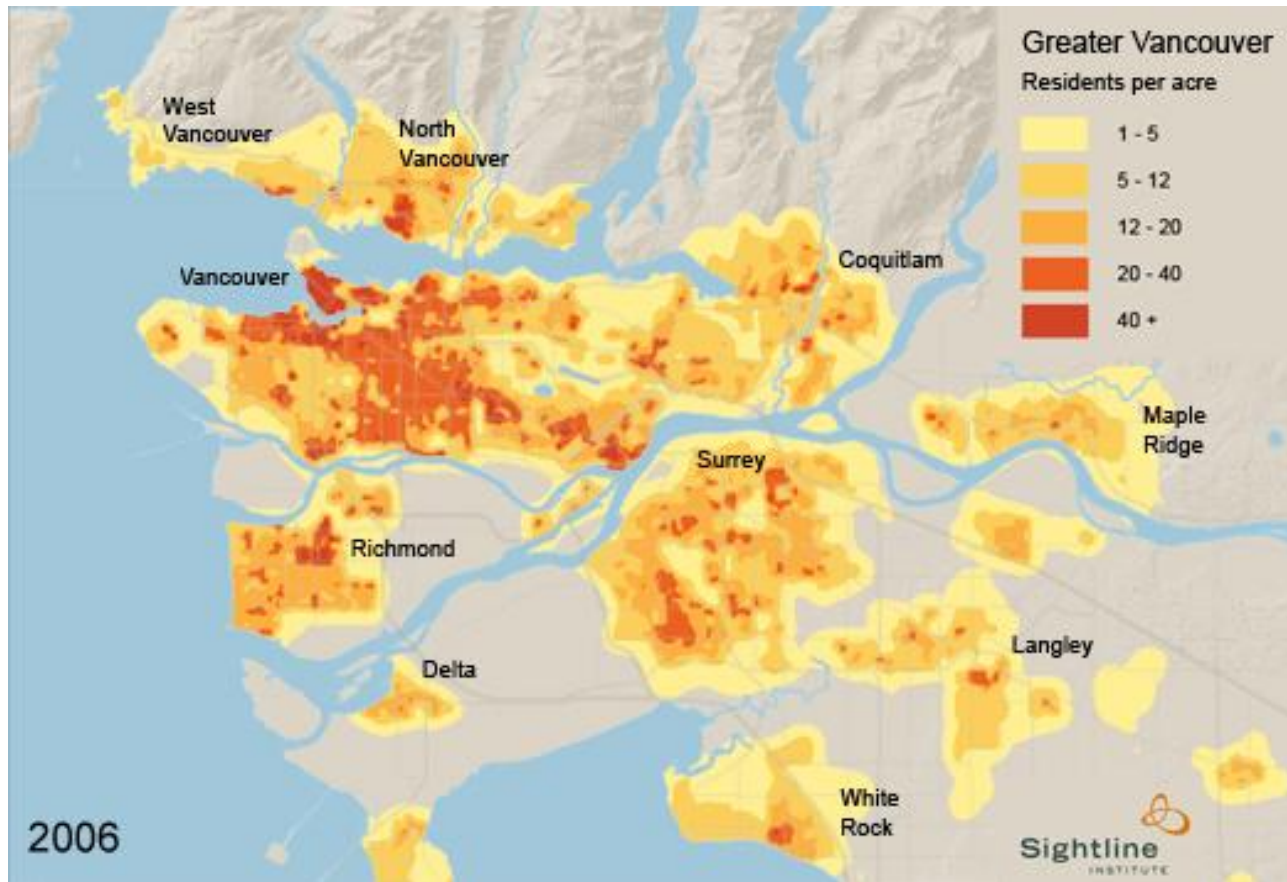
<http://www.geomika.com/blog/2011/03/12/tsunami-on-the-west-coast/>

Population Density in BC



https://www.reddit.com/r/MapPorn/comments/2n8sv4/population_density_map_of_british_columbia_3125/

BC Lower Mainland Population Density



<https://pricetags.files.wordpress.com/2008/05/sightline-map-lower-mainland.gif>

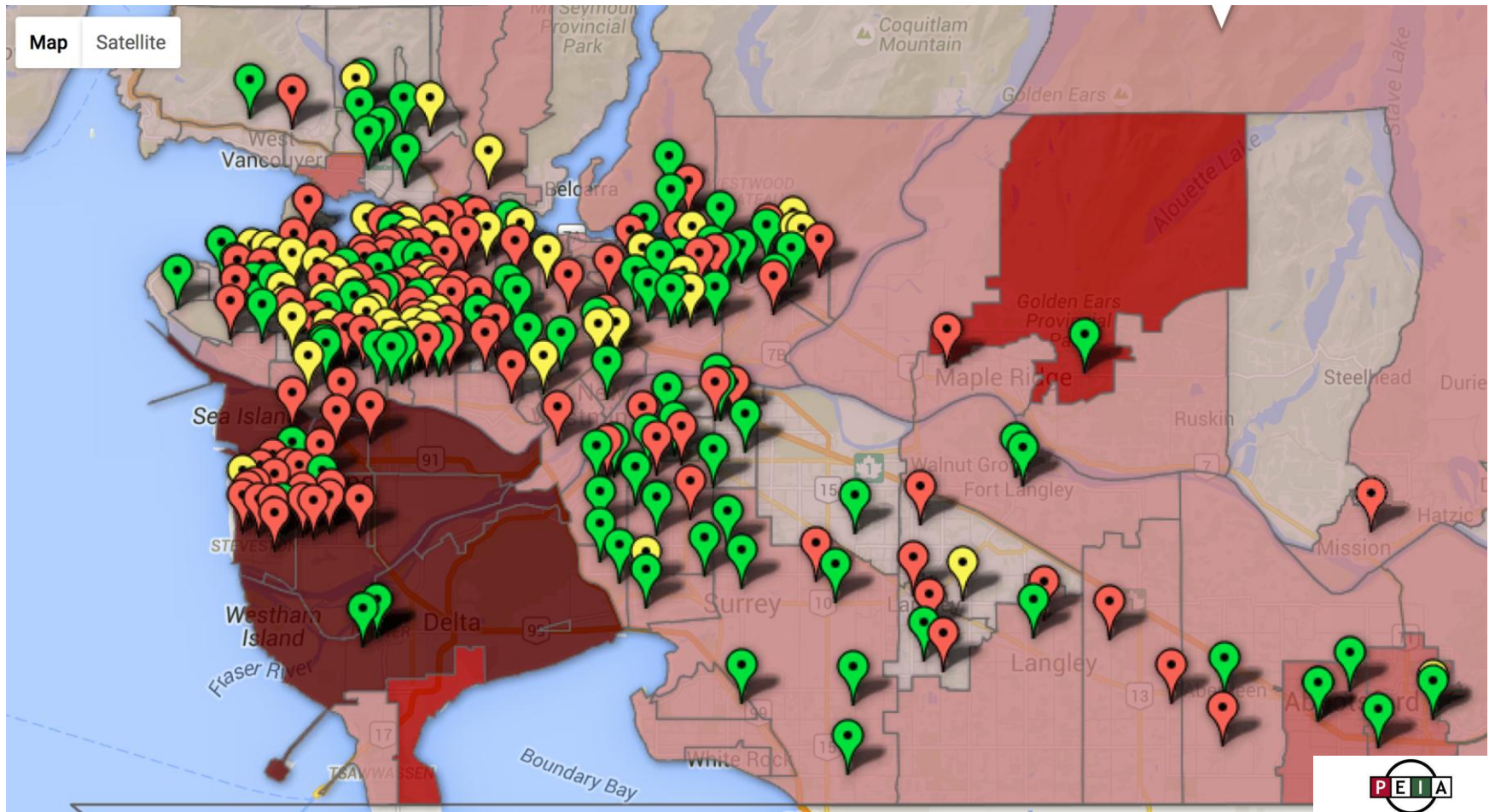
November 4, 2015

BC Earthquake Immediate Response

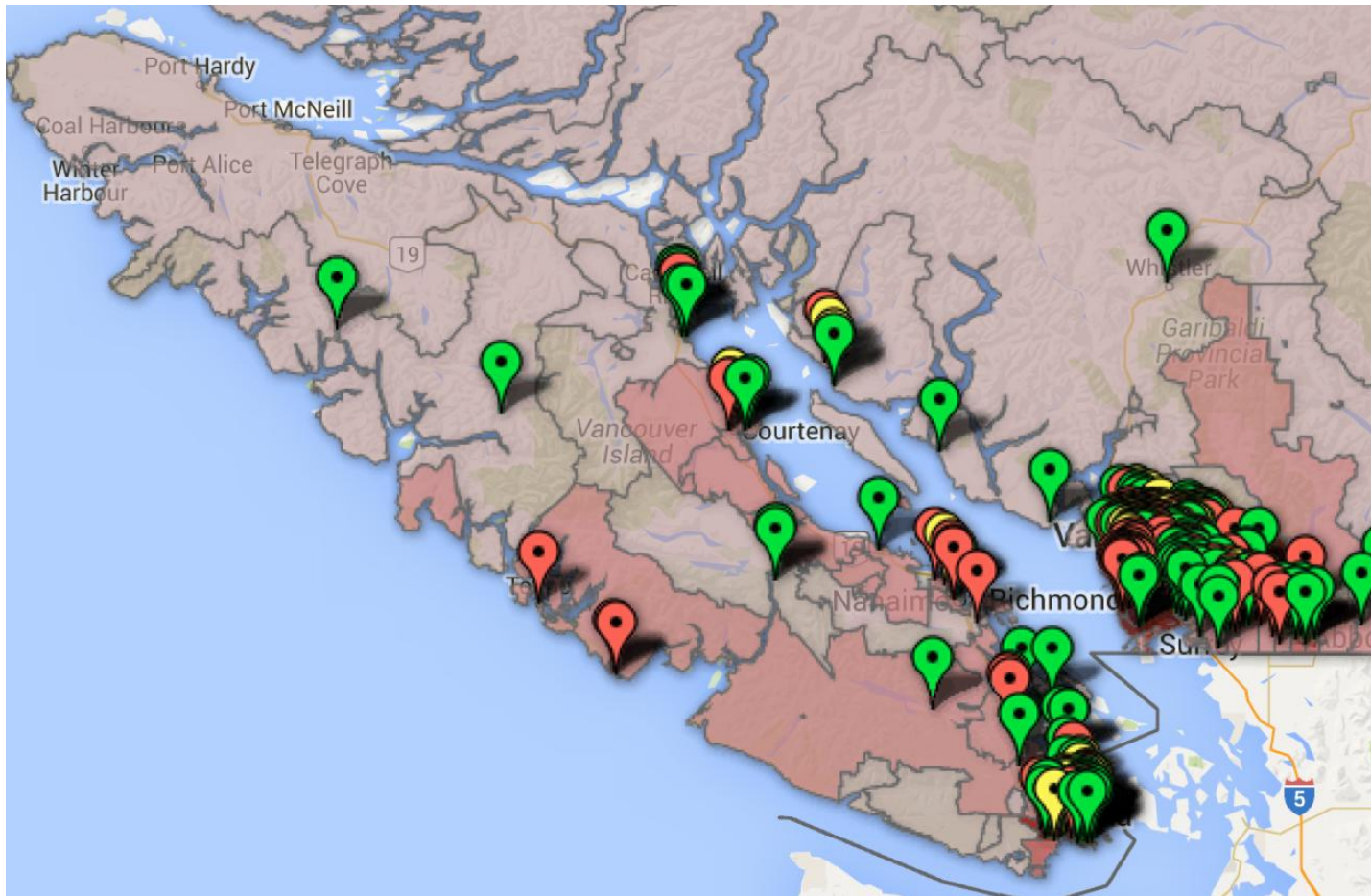
Building

- Modern buildings with inclusion of seismic constraints and industrial steel framed buildings prove the most resistant.
- VAN: 18 per cent of buildings in Metro Vancouver are most likely to receive extensive damage
- 12 per cent are most likely to receive complete damage.
- VIC: 18 per cent of buildings in the
- CRD are most likely to receive extensive damage and 11 per cent are most likely to receive complete damage.

Schools Seismic Risk 1



Schools Seismic Risk 2



Hospitals

Major buildings

- St. Paul's ER (Almost 100 years old)
- Vancouver General Hospital (VGH) Central Pavilion built 1959
- VGH Emergency Department Bldg (1978)
- Major Buildings at UBC Hospital, Richmond General Hospital, Lions Gate Hospital, Burnaby General Hospital, Langley Memorial Hospital, the B.C. Children's Hospital and the B.C. Women's Hospital have also been flagged at the highest earthquake risk in the reports.
- Non-Structural damage (equipment, gas tanks, pipe lines etc. may suffer significant damage)



PACIFIC ENERGY INNOVATION ASSOCIATION

Police Departments

- Most stations were build meeting code requirements



<http://www.timescolonist.com/news/preparing-for-a-disaster-1.10647>

Fire Halls and Ambulance Service Buildings

Most firehalls were built meeting latest code requirements



PACIFIC ENERGY INNOVATION ASSOCIATION

Vancouver Dedicated Fire Protection System

- Excellent fire water supply system
- Completed in 1995



<http://firepump.tripod.com/>



Fire Following Earthquake, edited by Charles Scawthorn, John M. Eidinger, Anshel Schiff
<http://spacing.ca/vancouver/2014/11/10/doors-open-vancouver-false-creek-pump-station/>

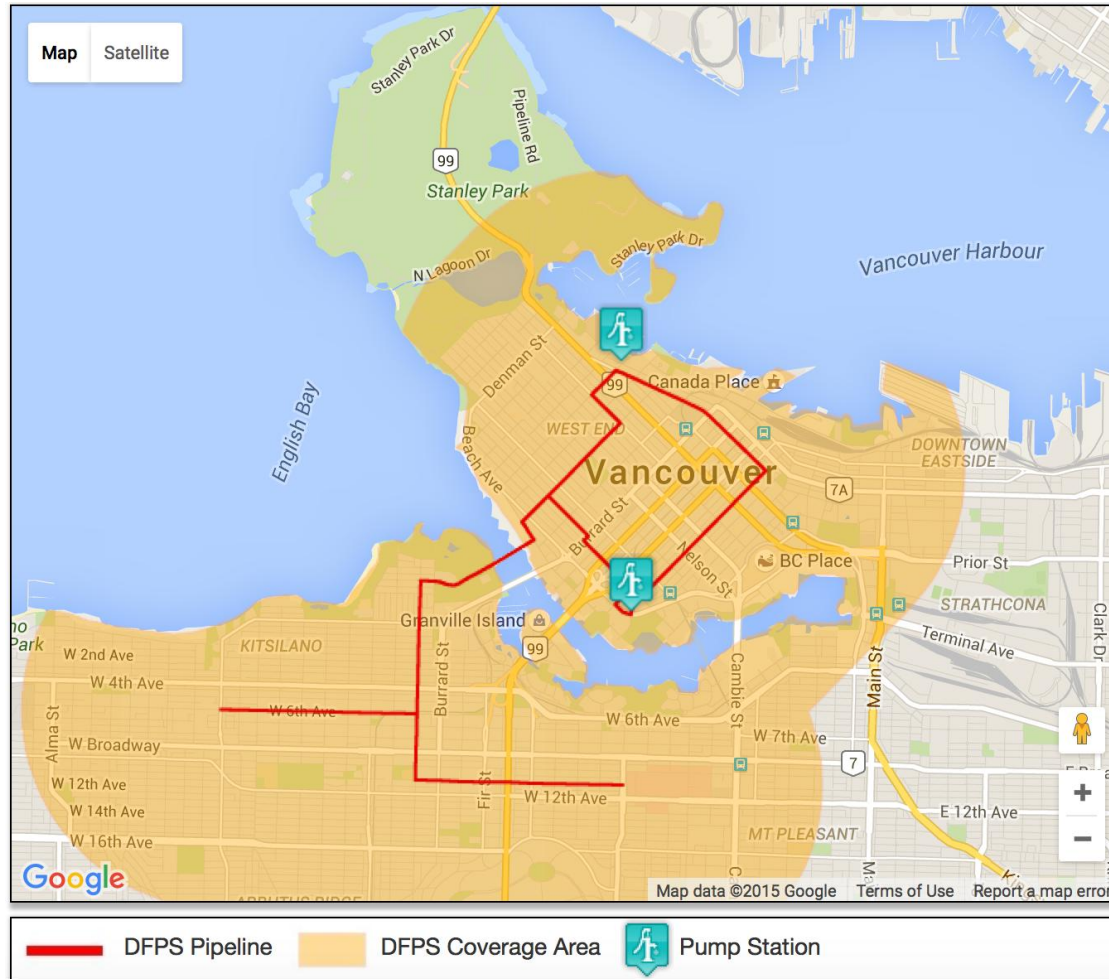
https://www.google.ca/search?q=city+of+vancouver+dedicated+fire+protection+system&biw=1526&bih=890&source=lnms&tbn=0CAcQ_AUoAigKahUKEwjY9qzflPLIAhVPlIgKHb-SAag#imgsrc=5MqLUddwcQoo4M%3A

November 4, 2015



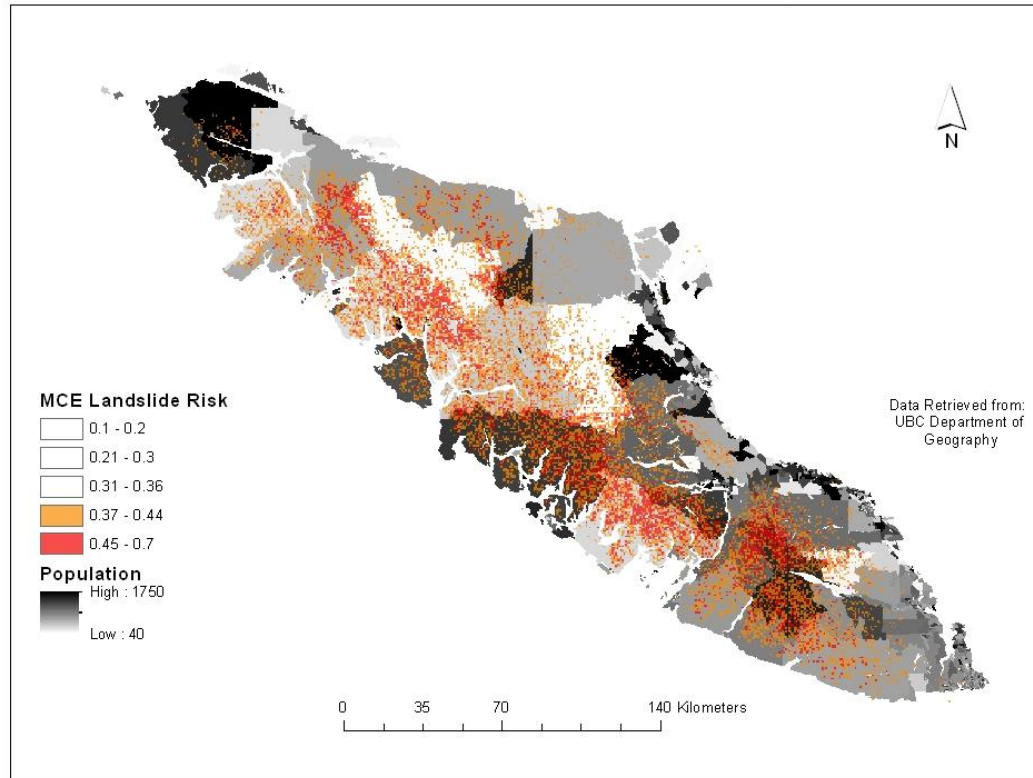
PACIFIC ENERGY INNOVATION ASSOCIATION

Vancouver Dedicated Fire Protection System



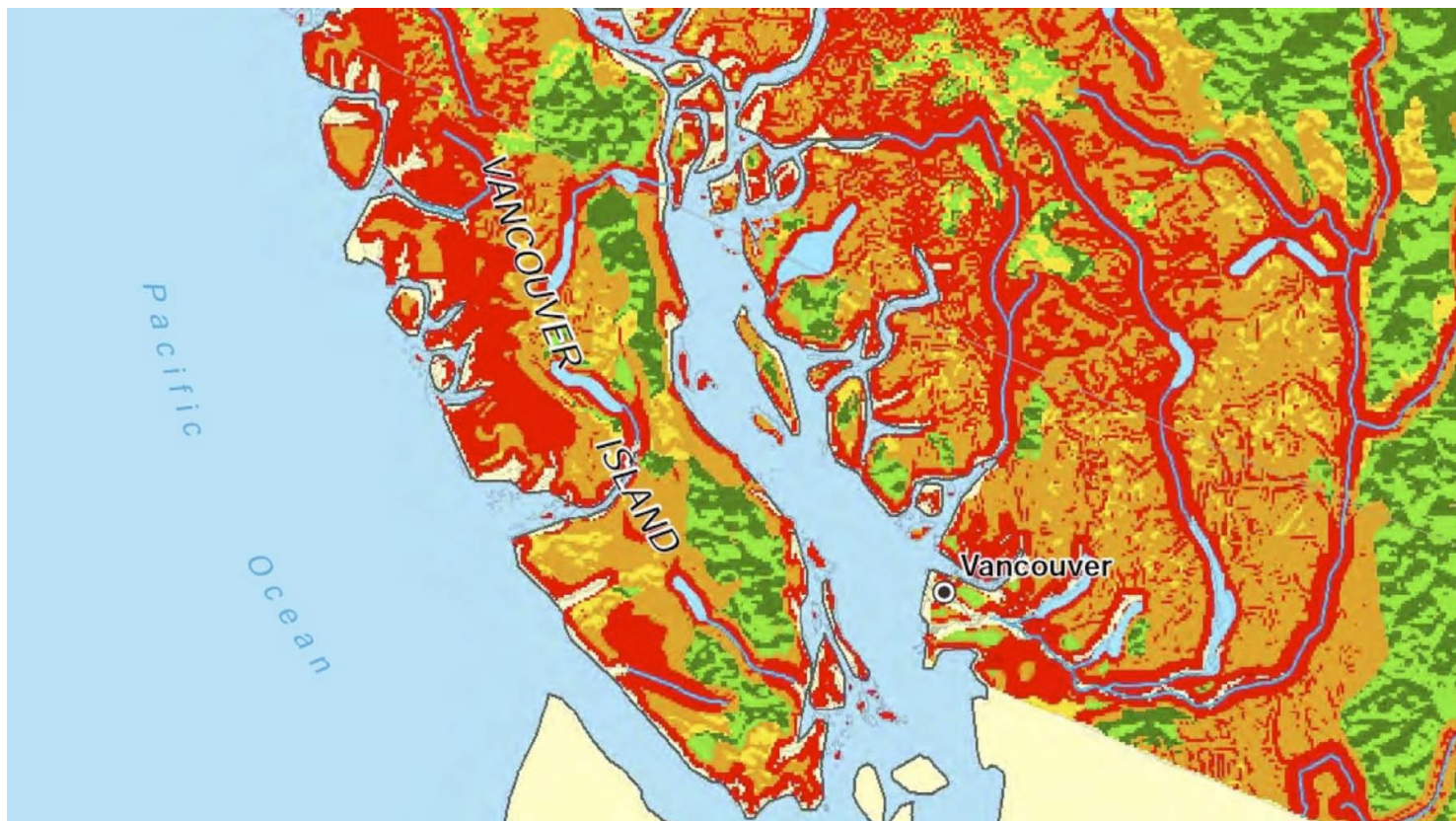
Landslide Risk Vancouver Island

Vancouver Island Landslide Risk Assessment: Population and Risk



http://ibis.geog.ubc.ca/courses/geob370/students/class11/astocks/www/images/pop_risk.jpg

Landslide Susceptibility



Land Slides



<http://ibis.geog.ubc.ca/courses/geob370/students/class11/astocks/www/intro.html>

Landslides



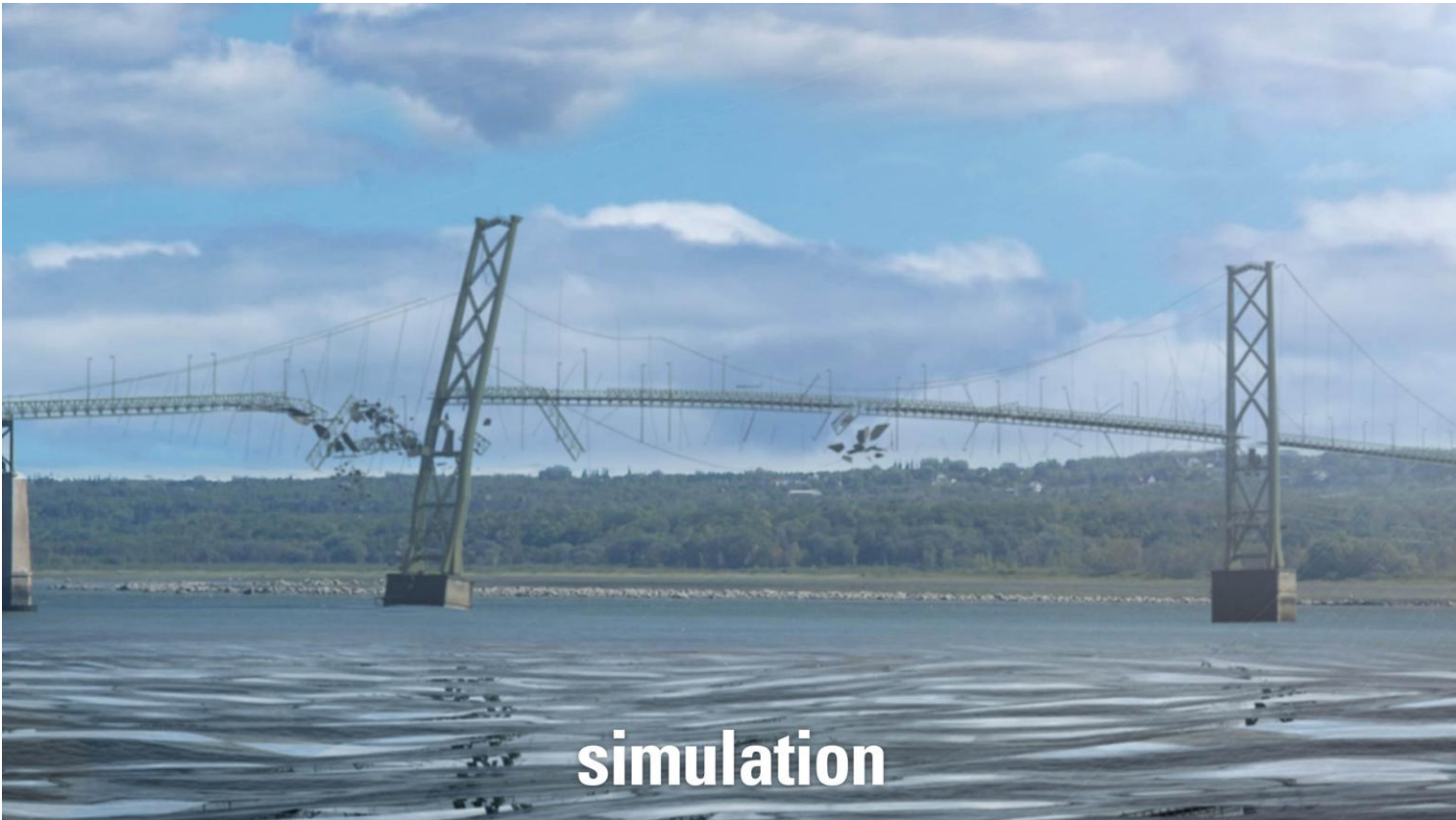
<https://www.collectionscanada.gc.ca/obj/021018/f1/xx010609-v6.jpg>

Roads



PACIFIC ENERGY INNOVATION ASSOCIATION

Bridges



<http://www.canadianunderwriter.ca/MktgImages/cu-ibc-quake/201310291351020me1est55122fpv4521wqk5id.jpg>

Bridges at Lower Mainland



<http://www.sfu.ca/geog/geog351fall12/project15/>

November 4, 2015



PACIFIC ENERGY INNOVATION ASSOCIATION

Bridges

- Kevin Baskin, chief bridge engineer for the Ministry of Transportation and Infrastructure, said the upgrades are designed to "minimize the risk of collapse" of key transportation infrastructures in an earthquake.
- The upgrades are designed for an earthquake measuring to 6.5 to 7.0 on the Richter scale, once in every 475 years.

<http://www.canada.com/closeup+vancouver+disaster+preparedness/1082327/story.html>

https://www.th.gov.bc.ca/publications/eng_publications/bridge/standards/MoT_SRDC.pdf



PACIFIC ENERGY INNOVATION ASSOCIATION

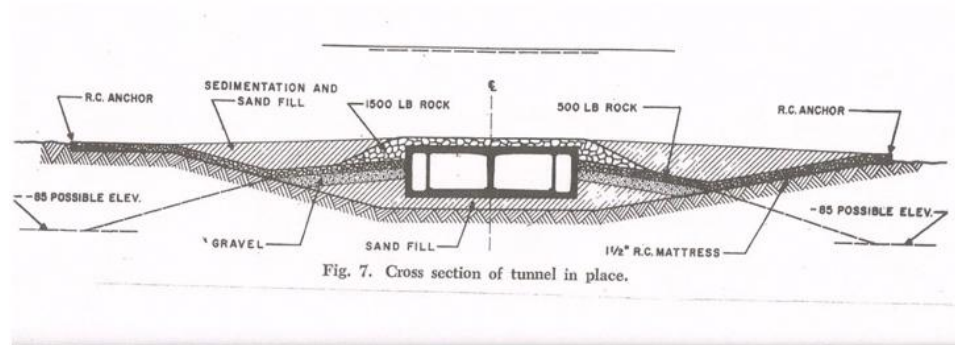
Port Mann Bridge



<http://www.abbynews.com/news/118198244.html>
November 4, 2015

George Massey Tunnel

- Built in 1959
- Design had very little consideration to earthquakes
- Built on liquefiable soil area
- Seismic retrofit completed in 2006 for 1 in 100 earthquake
- Earthquake early warning system was installed in 2009



<http://tranbc.ca/2012/07/23/emergency-what-if-scenario-are-highways-prepared-for-earthquakes/>

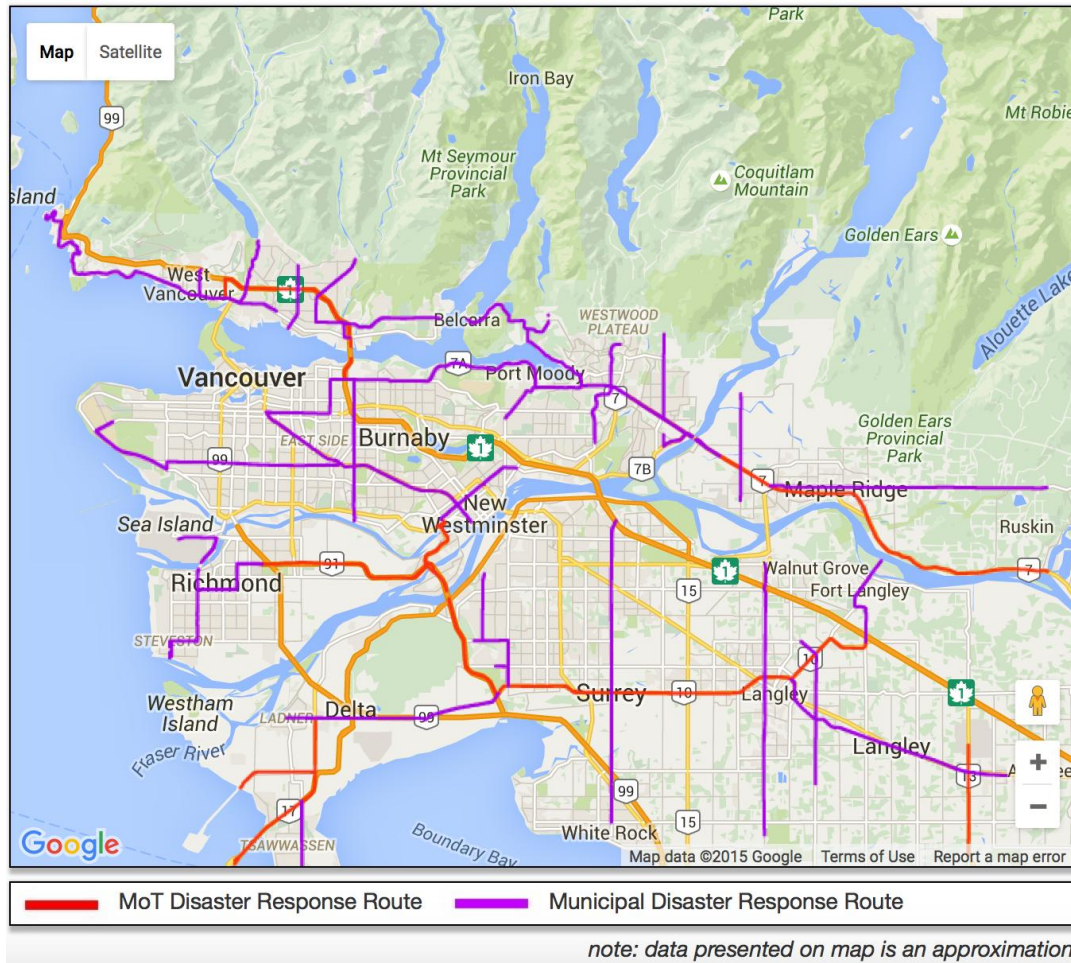
230 kV Line in George Massey Tunnel

- Tunnel upgrade was for 1 in 100 year event



<http://vancouver.24hrs.ca/2015/07/13/bc-hydro-reviews-tunnel-project-options>

Lower Mainland Disaster Response Routes



<http://www.sfu.ca/geog/geog351fall12/project15/>

Ferry Terminals Lower Mainland



BC Ferries Earthquake

- Vancouver Island, Sunshine Coast cut off
- Terminals are vulnerable to earthquakes
- No Seismic upgrade program in place



<http://www.cbc.ca/news/canada/british-columbia/megaquake-would-crumble-bc-ferries-terminals-says-engineer-ray-hebden-1.25>

Airports at Lower Mainland



Ports at Lower Mainland



<http://www.sfu.ca/geog/geog351fall12/project15/>

November 4, 2015

Railways at Lower Mainland



<http://www.sfu.ca/geog/geog351fall12/project15/>

November 4, 2015

Railways



<https://i.guim.co.uk/img/static/sys-images/Guardian/Pix/pictures/2007/07/16/train372.jpg?w=620&q=85&auto=format&sharp=10&s=315b7b94f2cd2ac26bf737a9f1f5>

November 4, 2015



PACIFIC ENERGY INNOVATION ASSOCIATION

Water Supply System

- Large portion of the water supply (about 77%) is seismically vulnerable
- Mainly due to soil liquefaction
- Cast iron pipes are in service
- Upgrade in progress addressing major issues

Richmond Water Supply

- After a 1 in 100 year event some water
- Unlikely any water after a 1 in 475 year event

http://www.richmond.ca/__shared/assets/072803_item72986.pdf



PACIFIC ENERGY INNOVATION ASSOCIATION

Telephone, Telus

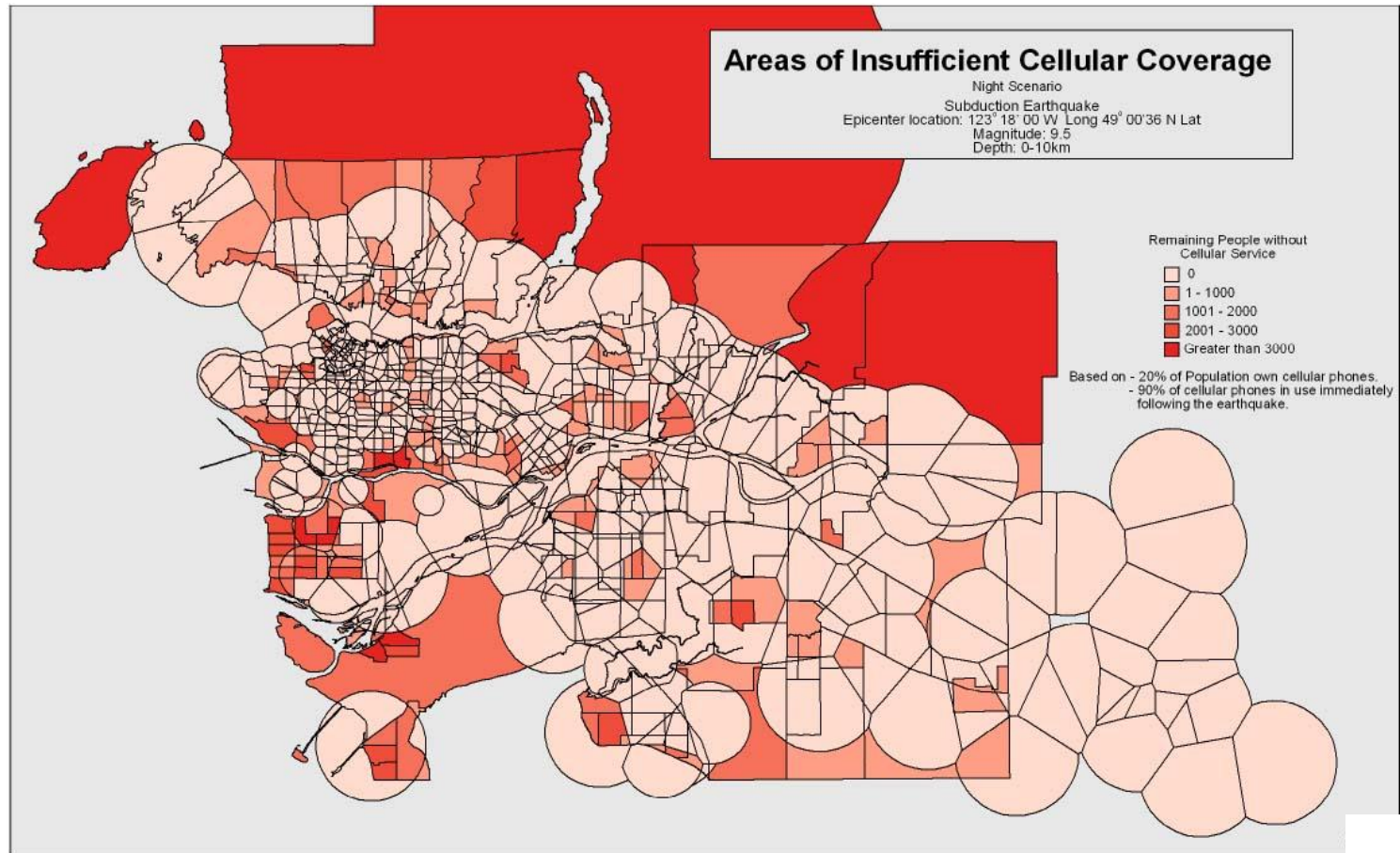
- undertaken an extensive seismic upgrade program focusing on major buildings in areas that are considered higher risk earthquake zones.
- Most major equipment buildings are classed as post-disaster structures because telephone service is essential in a disaster or emergency.
- Switches are braced for an earthquake with uninterrupted power supply and stand-by power in the form of generators and batteries.

https://about.telus.com/community/english/about_us/for_our_customers/customer_safety/disaster_planning/preparing_our_facili



PACIFIC ENERGY INNOVATION ASSOCIATION

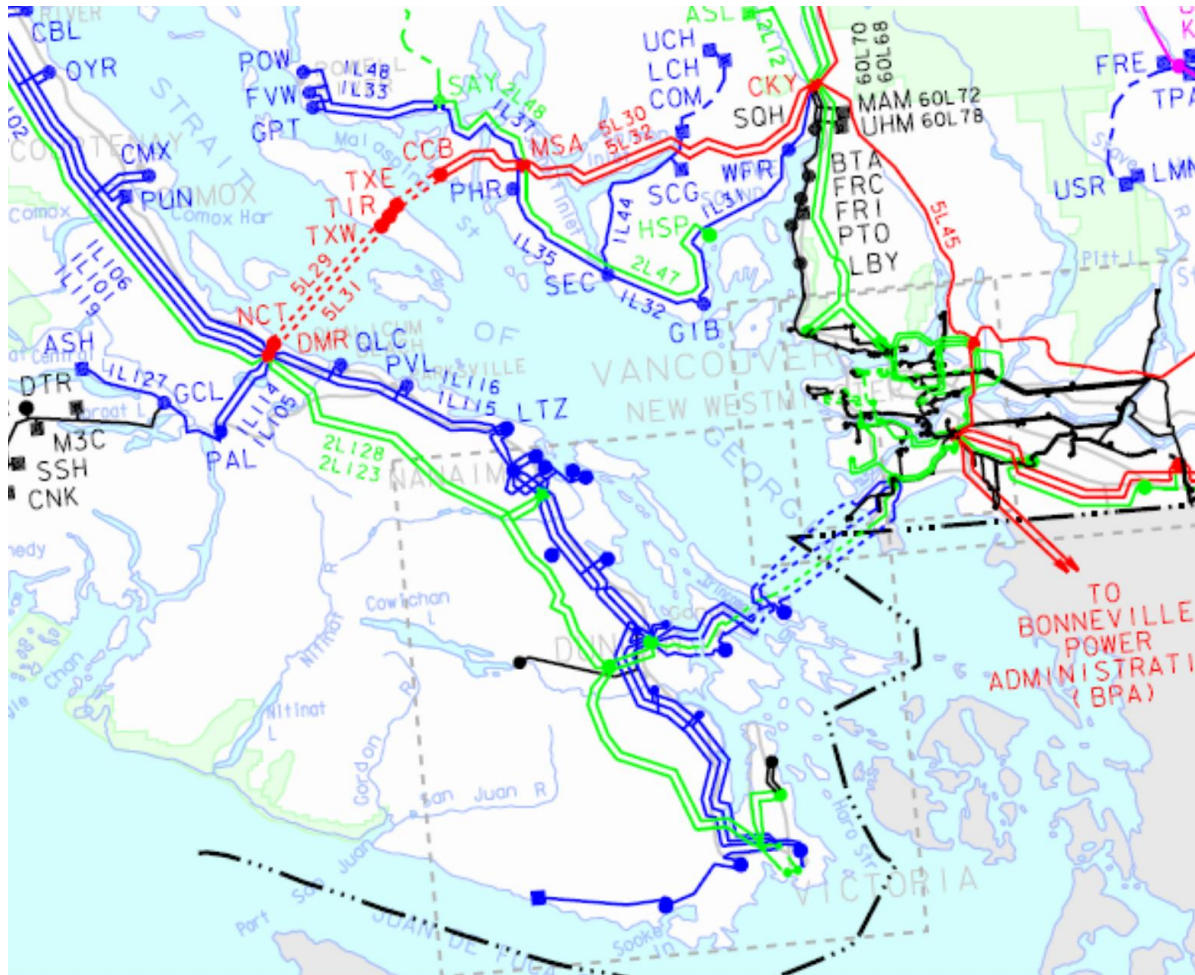
Phone System



Electrical System Major Vulnerabilities

- High Voltage Circuits and Submarine Cables
- Power Transformer Bushings
- Pot Heads
- Tsunami Inundation of Substations
- Transmission Towers

Vancouver Island Lower Mainland Electrical Transmission Ties

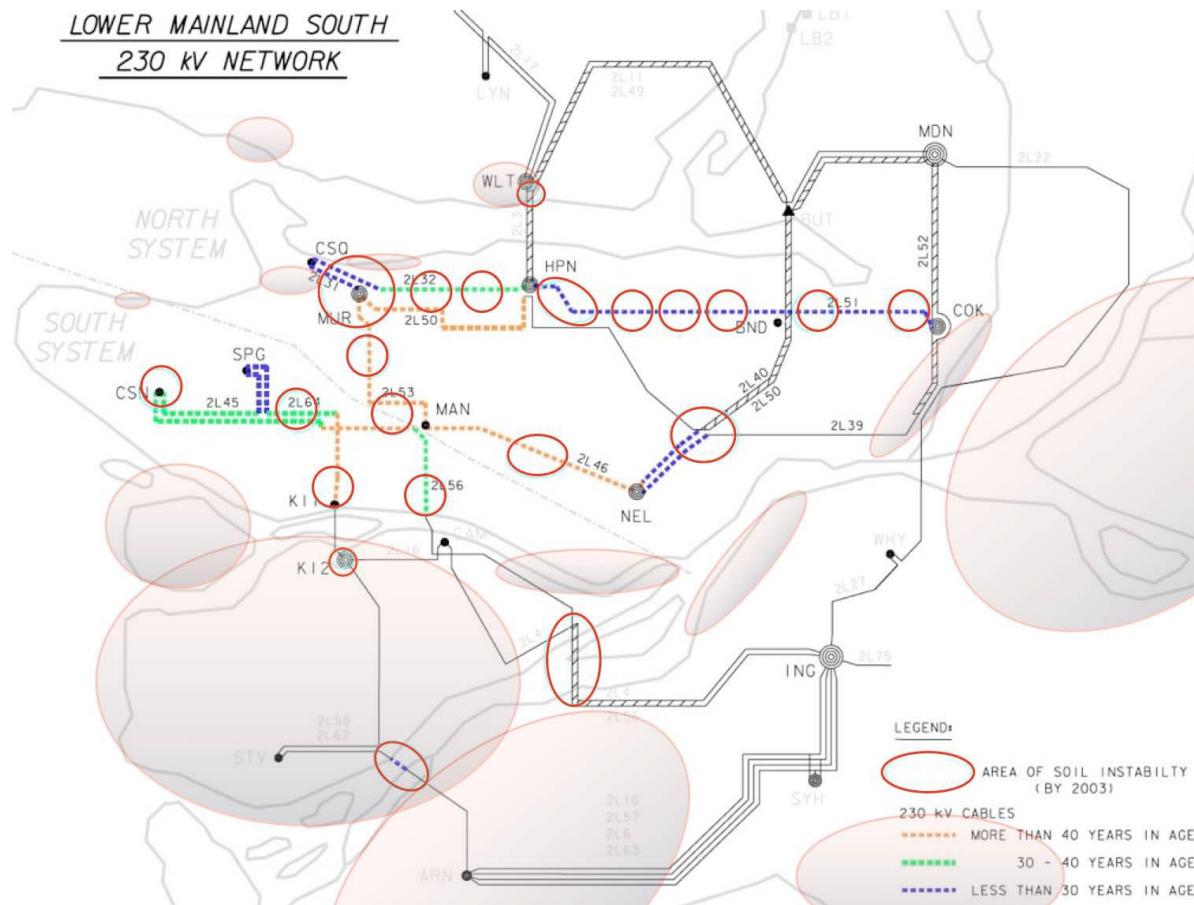


Pot Head Damage

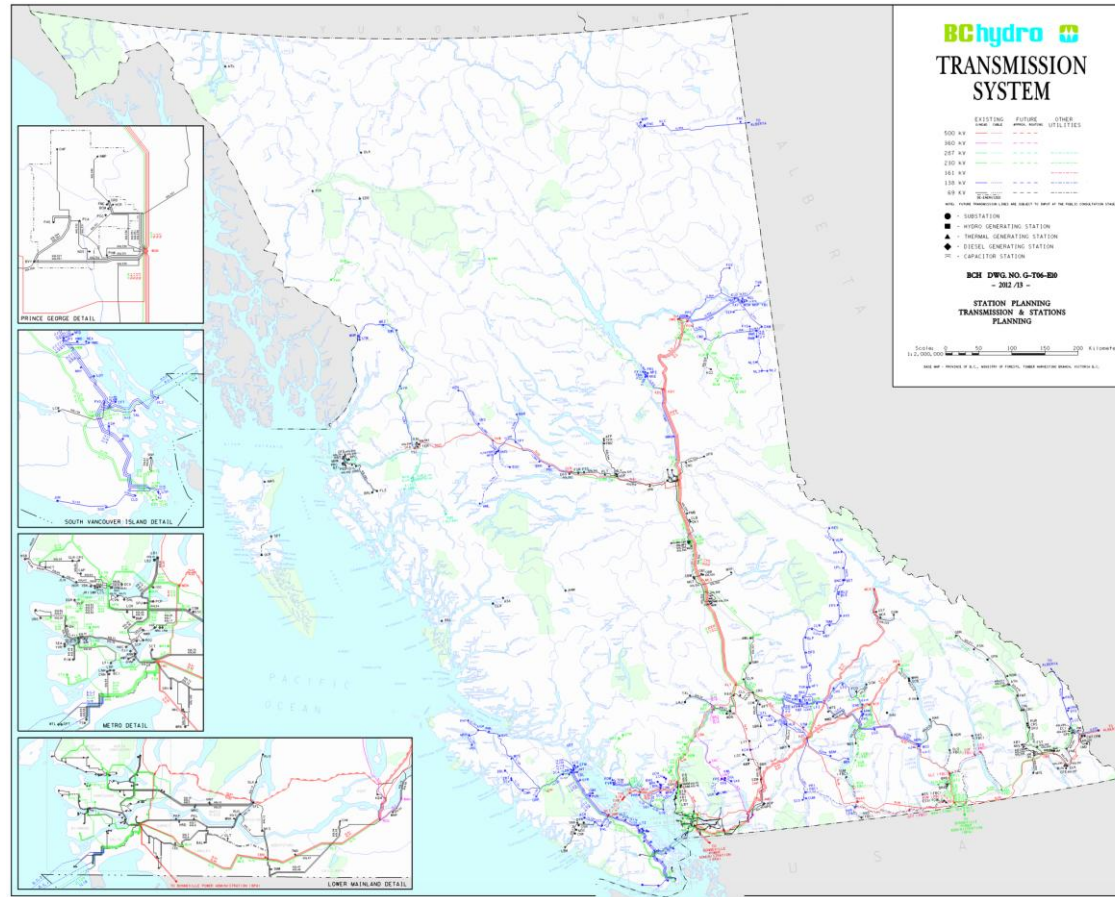
- 230 kV supply to Lower Mainland (Possible recovery time between
- 2 years to 4 years)
- 500 kV cable supply to Vancouver Island Possible recovery time between
- (2.5 years to 6 years)



Liquefaction Hazard Electrical Transmission, Lower Mainland



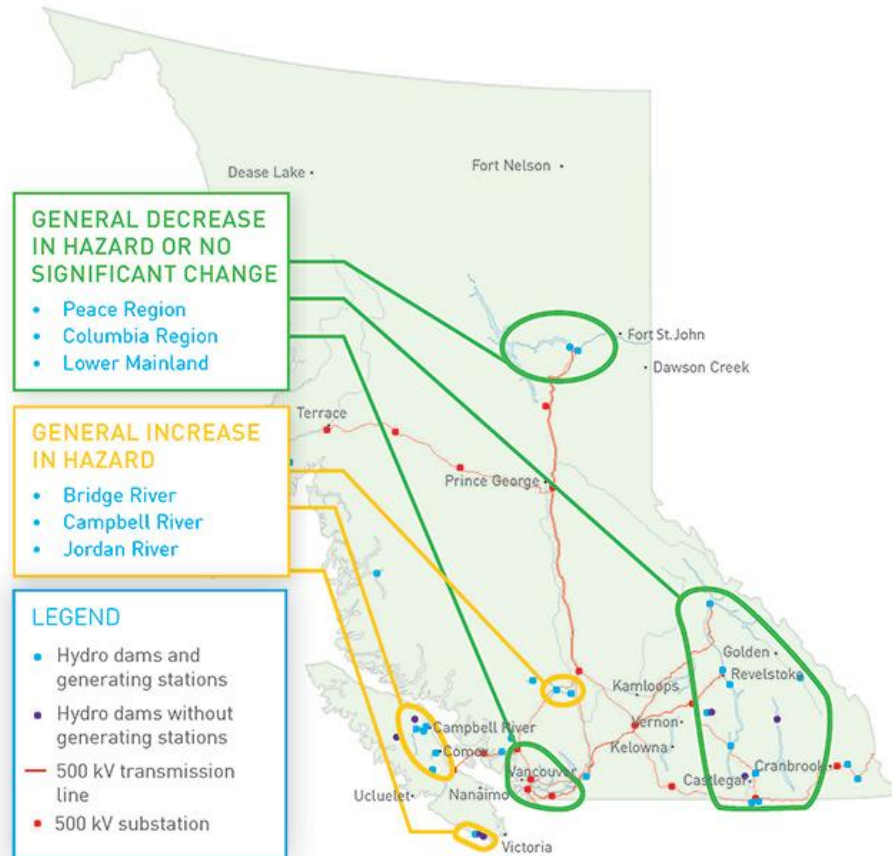
BC Hydro Transmission System



<https://www.bchydro.com/content/dam/BCHydro/customer-portal/documents/accounts-billing/electrical-connections/bchydro-systems-map.pdf>

BC Hydro Generation

CHANGE IN SEISMIC HAZARD



Revelstoke Dam



<https://multco.us/file/12819/download>

<http://www.revelstokereview.com/news/118929699.html>

http://www.bcuc.com/documents/proceedings/2006/doc_13359_b-40_bch_undertaking.pdf

November 4, 2015



PACIFIC ENERGY INNOVATION ASSOCIATION

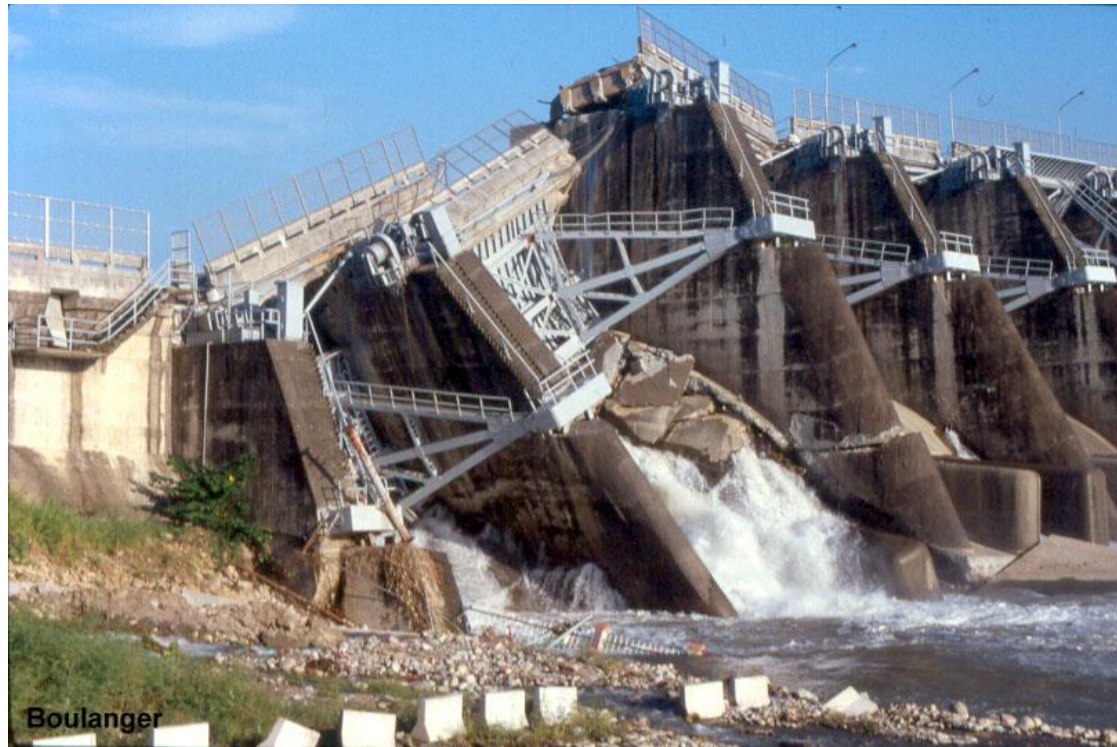
Columbia River Dams



<https://multco.us/file/12819/download>

<https://www.energy-northwest.com/ourenergyprojects/Columbia/PublishingImages/Columbia-Dams.jpg>

Earthquake Damaged Hydro Dam, Asia



South Asian Disaster Knowledge Network
http://www.saarc-sadkn.org/can_we.aspx

November 4, 2015

BC Hydro Lower Mainland Transmission



<https://www.bchydro.com/content/dam/BCHydro/customer-portal/documents/accounts-billing/electrical-connections/bchydro-transmission-systems-map.pdf>

Mount Pleasant Substation



<https://www.bchydro.com/news/conservation/2014/mount-pleasant-substation.html>
November 4, 2015



PACIFIC ENERGY INNOVATION ASSOCIATION

Control Centre

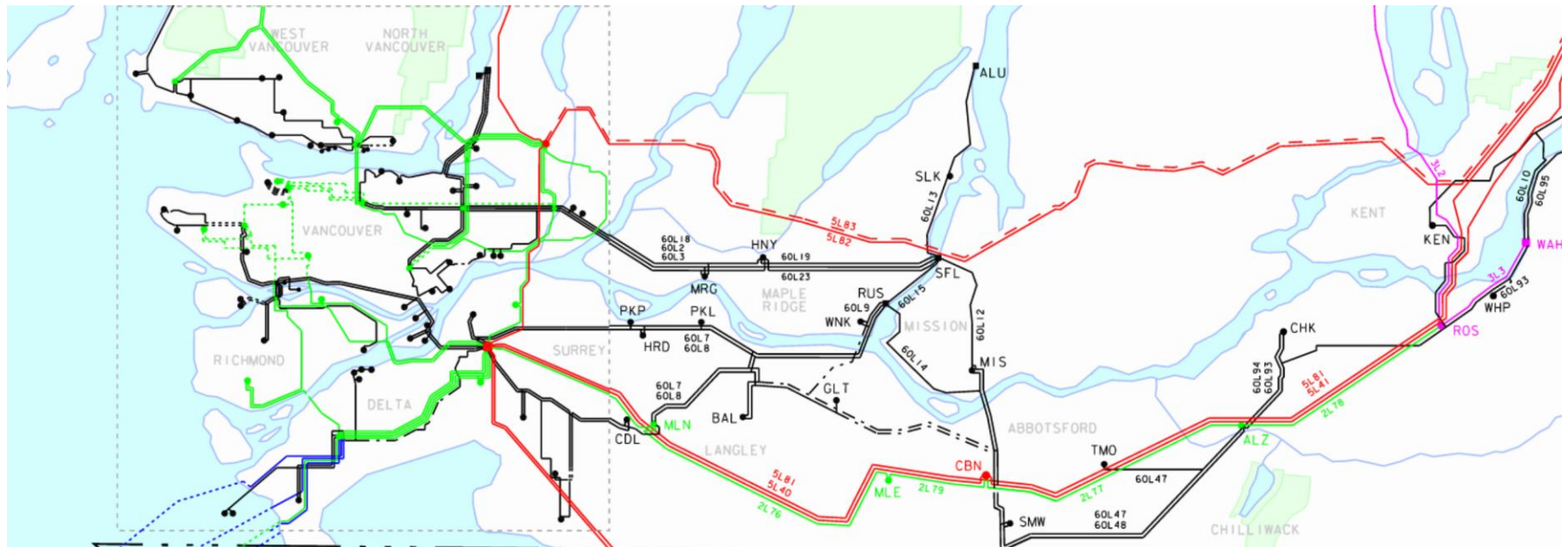
- The Fraser Valley Operations Centre is a
- An additional
- facility in the Okanagan is also available to ensure BC Hydro can instantly transfer control should anything happen to the primary centre.
- This improves the grid's reliability and the ability to withstand a major seismic event in the lower mainland.

https://www.bchydro.com/content/dam/hydro/medialib/internet/documents/safety/pdf/bc_hydro_seismic_event.pdf



PACIFIC ENERGY INNOVATION ASSOCIATION

BC Hydro Lower Mainland Transmission



<https://www.bchydro.com/content/dam/BCHydro/customer-portal/documents/accounts-billing/electrical-connections/bchydro-systems-map.pdf>

Pothhead damage



<http://www.exponent.com/files/Uploads/Images/electrical/Earthquake%20damagethumb.jpg>

Transformers



November 4, 2015

Towers



Japan



<http://earthquake.usgs.gov/learn/glossary/images/lifelines.jpg>

November 4, 2015

Auditor General of BC on BC's Earthquake Preparedness



OFFICE OF THE
Auditor General
of British Columbia

8 Bastion Square
Victoria, British Columbia
Canada V8V 1X4
Telephone: 250-419-6100
Facsimile: 250-387-1230
Website: www.bcauditor.com

The Honourable Linda Reid
Speaker of the Legislative Assembly
Province of British Columbia
Parliament Buildings
Victoria, British Columbia
V8V 1X4

Dear Madame Speaker:

I have the honour to transmit to the Legislative Assembly of British Columbia my *Catastrophic Earthquake Preparedness* report.

We conducted the audit in accordance with Section 11(8) of the *Auditor General Act* and the standards for assurance engagement established by the Canadian Institute of Chartered Accountants.

My Office first looked at this topic 17 years ago in our *Earthquake Preparedness* report and arrived at the same conclusion. Progress has been made in some areas over this intervening period, but overall, the province is still at a significant risk if a catastrophic earthquake were to occur today.

One of the main causes for this lack of progress is that preparing for a catastrophic earthquake has not been made a priority. Since my Office's report in 1997, successive governments have decided to allocate scarce public resources to meet more immediate pressing demands, rather than to adequately prepare the province for a catastrophic earthquake that may or may not occur. As a result, Emergency Management BC (EMBC), the government organization tasked with leading the Province's response to a catastrophic earthquake, has not fulfilled this part of its mandate.

We concluded that EMBC cannot demonstrate that it is adequately prepared to manage the effects of a catastrophic earthquake and it is not reporting publicly on the Province's preparedness.

While the recommendations from this audit are directed at EMBC, I urge all readers of this report to look at their own situation and ask themselves whether they are ready and if they have done everything they should to prepare themselves, their families, and their communities for an earthquake.

I would also like to take this opportunity to express my appreciation to EMBC and Ministry of Justice staff for their cooperation during this audit.

Russ Jones, MBA, CA
Auditor General
Victoria, British Columbia
March 2014

“Since my Office’s report in 1997, successive governments have decided to allocate scarce public resources to meet more immediate pressing demands, rather than to adequately prepare the province for a catastrophic earthquake that may or may not occur. As a result, Emergency Management BC (EMBC), the government organization tasked with leading the Province’s response to a catastrophic earthquake, has not fulfilled this part of its mandate.”



PACIFIC ENERGY INNOVATION ASSOCIATION

Auditor General of BC, Catastrophic Earthquake Preparedness
www.bcauditor.com

November 4, 2015

Cycle of Earthquake Preparedness

- Analysis and assessment
- Remediation
- Indication and warning
- Mitigation
- Incident response
- Recovery and Reconstruction

Questions

